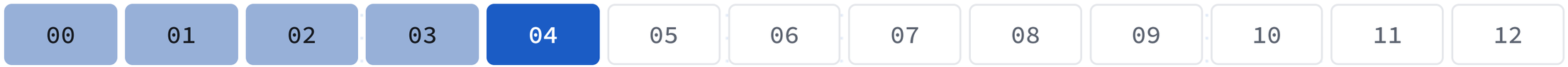
MODULE 04 · AMBITIOUS ENTREPRENEURSHIP

Validating Problems

What evidence would change your mind?

COURSE PATH



COURSE MATERIAL · SCAN NOW

All slides live on GitHub

The newest version, free, no account needed

github.com/BaNburger/
AmbitiousEntrepreneurship

1. **Open the link**
   github.com/BaNburger/AmbitiousEntrepreneurship
2. **Go to modules**
   then 04-validating-problems
3. **Click the lecture PDF**
   M04_Lecture_Validating-Problems.pdf, download top right

Public course repository of Ambitious Entrepreneurship. The QR code opens github.com/BaNburger/AmbitiousEntrepreneurship.

WHERE WE ARE · THE COURSE

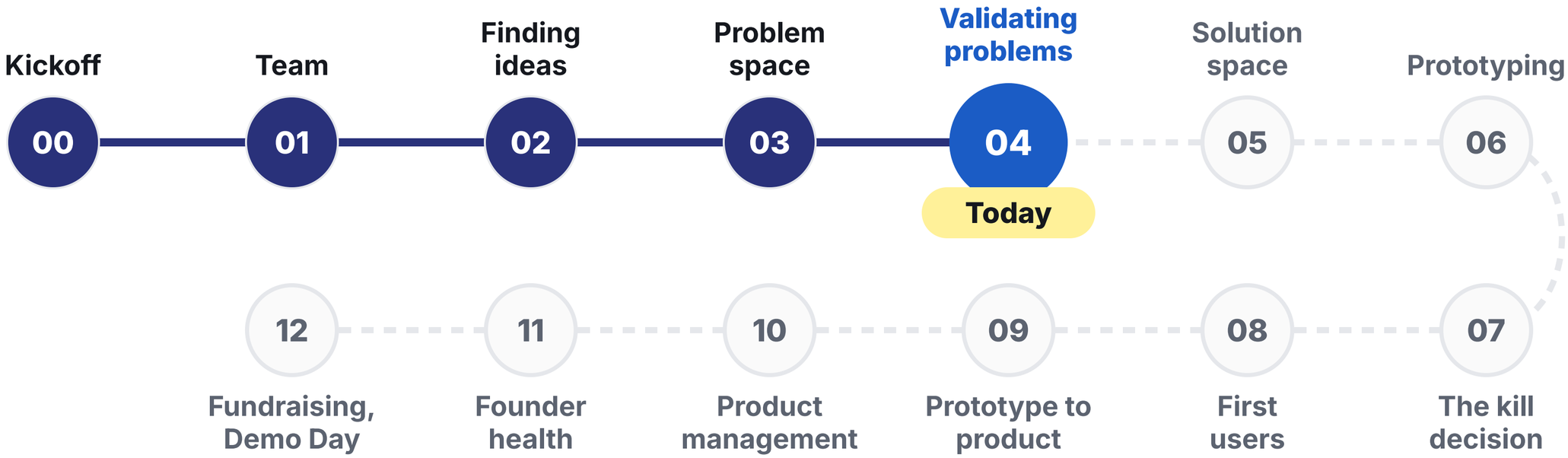
From what people say to what they do

Thirteen sessions from first idea to first revenue. What teams have by now:

One idea, picked on evidence
Finding ideas (M02)

A hair-on-fire segment
Problem space (M03)

Named people for a next step
Problem space (M03)



CHECK-IN · SLIDO OR HANDS

How many people agreed to a **next step**?

1

2

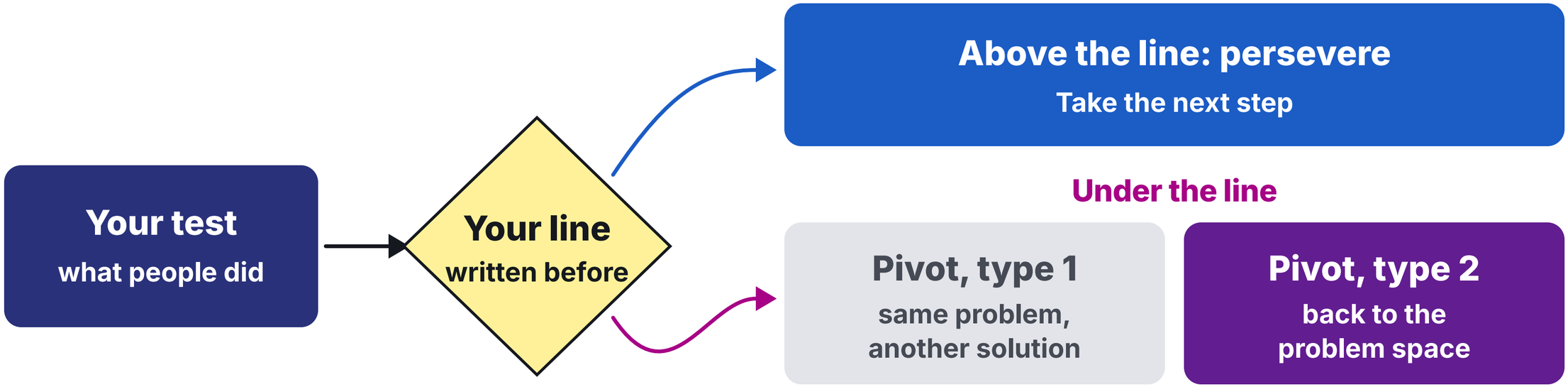
3

2:00

PIVOT OR PERSEVERE

Under your line? That is a decision, not a failure

Before there is a solution, a pivot goes back to the problem



Both are progress. The only miss: no line, so no decision

FRAMEWORK Pivot or persevere after Ries, E. (2011), The Lean Startup. Pivot types as in the course Glossary.

ACCOUNTABILITY

Three questions, every session

1

User interactions?

What did you learn?

2

New revenue?

Why did they pay?

3

Shipped?

Why does it matter?

🦄 **Report target vs. actual in small groups**

CHAPTER 1 OF 9

Previously on Ambitious Entrepreneurship

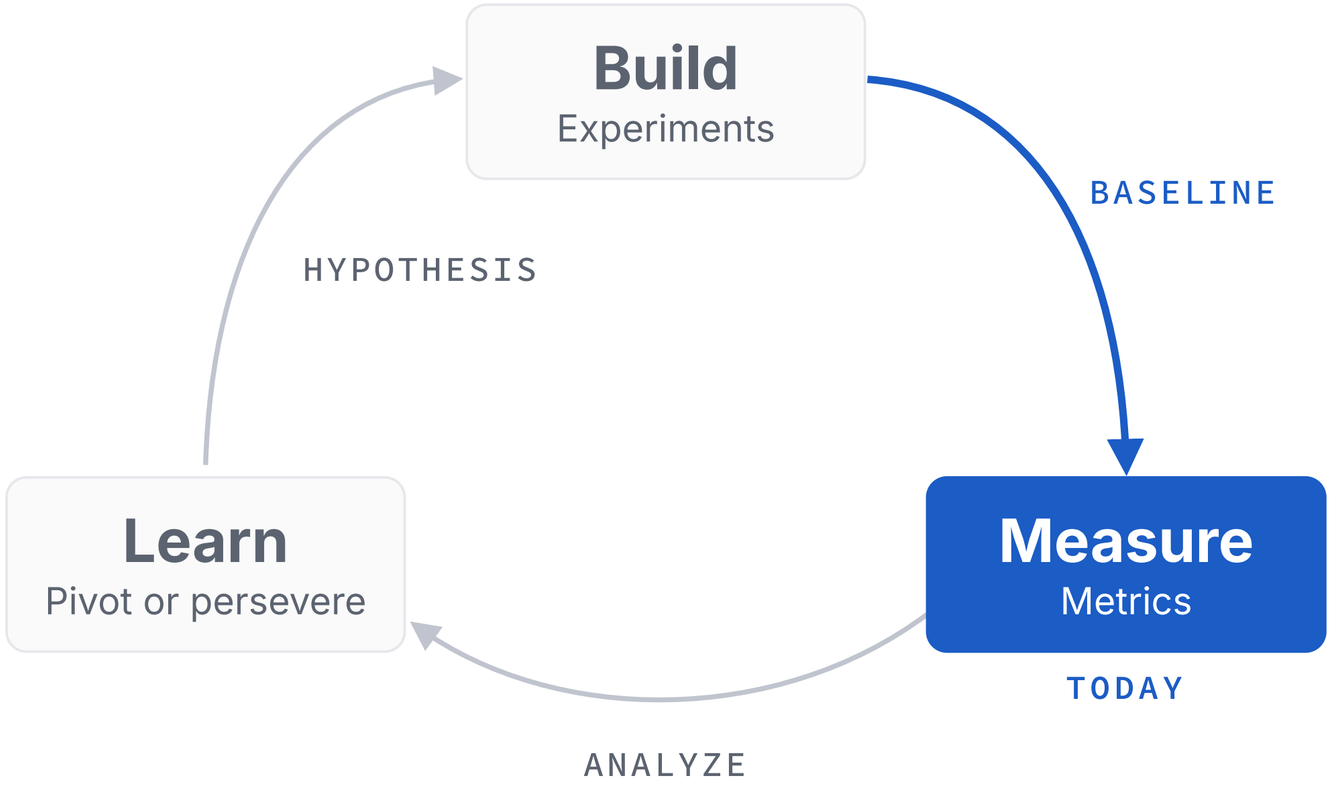
You heard who hurts, and from today you count who acts.

01 **Previously on Ambitious Entrepreneurship**
02 Your presentations
03 **Rank it: what your evidence proves**
04 **Draw it: the line before the data**
05 **Launch it: a page that tests your problem**
06 **Count it: visitors from your segment**
07 Wrap-up and check-out
08 Workshop: launch a page that tests your problem
09 Case discussion

PREVIOUSLY · KICKOFF (M00)

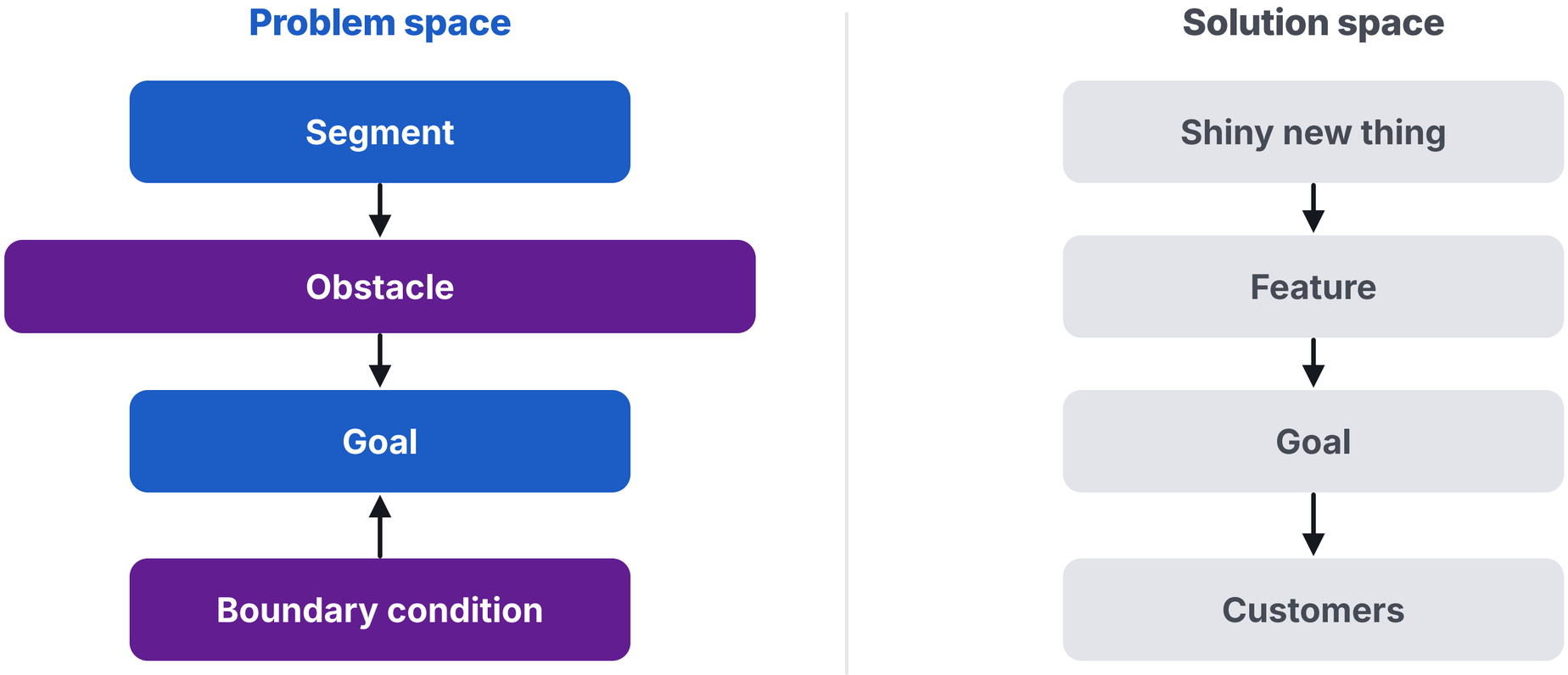
Your idea is a hypothesis

Today you measure what people do



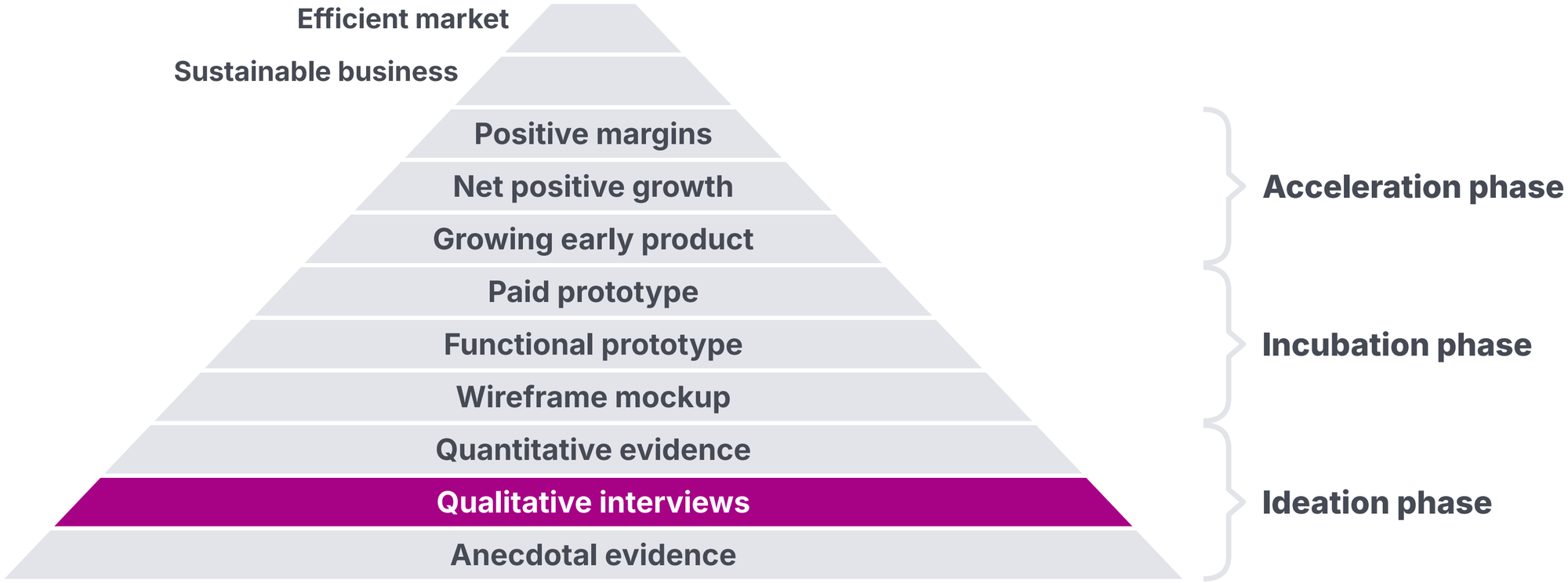
PREVIOUSLY · PROBLEM SPACE (M03)

You know who hurts. Now count who acts



PREVIOUSLY · PROBLEM SPACE (M03)

You stood on the second step



TODAY'S QUESTION · HANDS

What evidence would change your mind?

1

2

1:00

CHAPTER 2 OF 9

Your presentations

Three minutes per team to show what your records say since last time.

01 ~~Previously on Ambitious Entrepreneurship~~

02 **Your presentations**

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

YOUR PRESENTATIONS · MAKE IT CREDIBLE

Show us where you're at in 3 minutes or less!

3:00

Facts

Statistics or research

What does your segment spend?

Where is it from?

Numbers

Your Interview Log

Conversations held

Next steps agreed

Spend in € or hours

Quotes

Interviews

The three that decided it

Images

Screenshots or users

The workaround, if they showed it

Verdict: segment, job, stay or switch

Hypothesis: three people likely to pay first

What did you do?

What did you learn?

How can we help?

INPUT 1 OF 4 · CHAPTER 3 OF 9

Rank it: what your evidence proves

What you know is only as strong as the step it stands on.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

INPUT 1 OF 4 · RANK IT · THREE QUESTIONS

What is your evidence worth?

1

Which evidence is strongest?

2

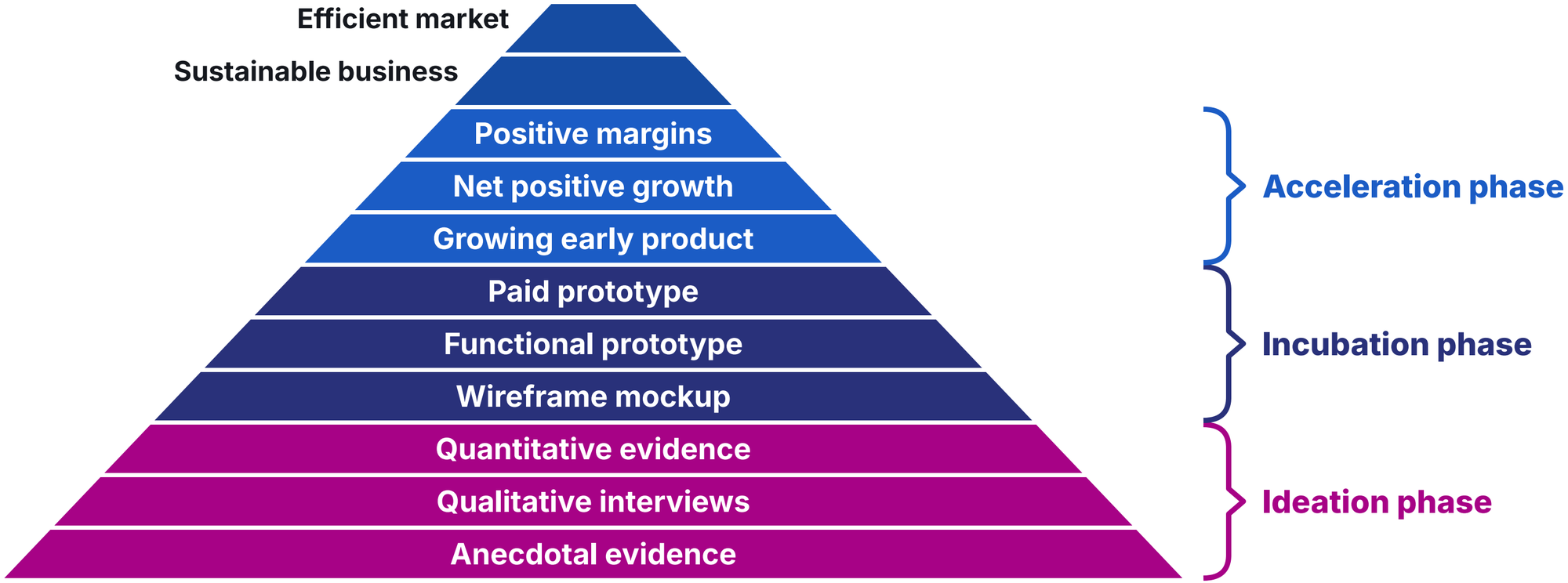
When does a survey count?

3

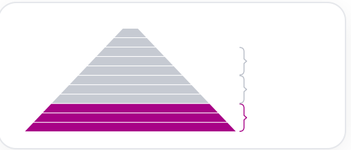
What validates a problem?

CORNERSTONE · HIERARCHY OF EVIDENCE

Climb from what they say to what they pay



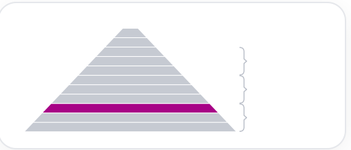
BASTIAN Hierarchy of evidence in the lean startup. A practitioner ladder, untested as a whole.



THE RUNNING EXAMPLE · THE ARM THAT HURTS

Anecdotes diagnose. Interviews theorize. Numbers verify

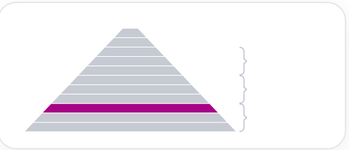
Anecdote	Interviews	Numbers
N under 5 · subjective	N under 20 · unstructured	N over 20 · structured
		
"When I do this, my arm hurts."	"Doing this is ok, but that hurts."	"Sore arm in 90% of runs against a pole."
Diagnose	**Theorize**	**Verify**



SURVEYS · WHEN THEY COUNT

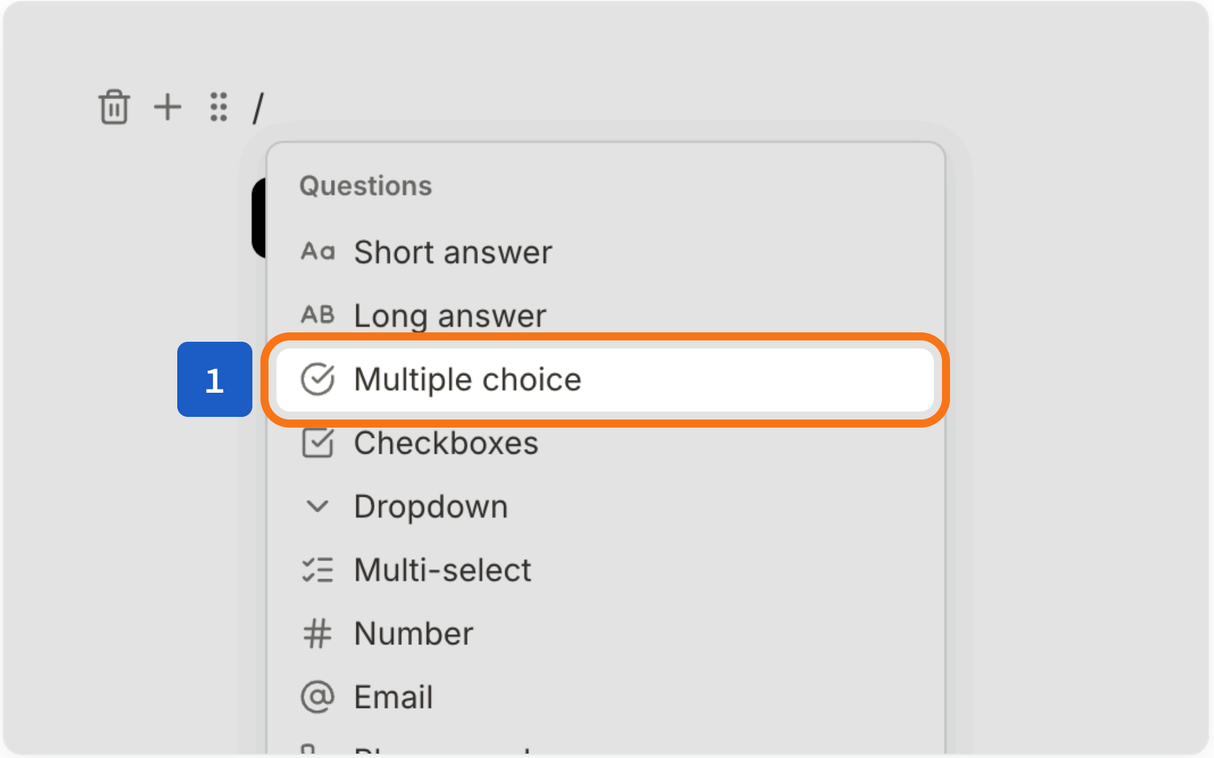
A survey counts only when it asks about the past

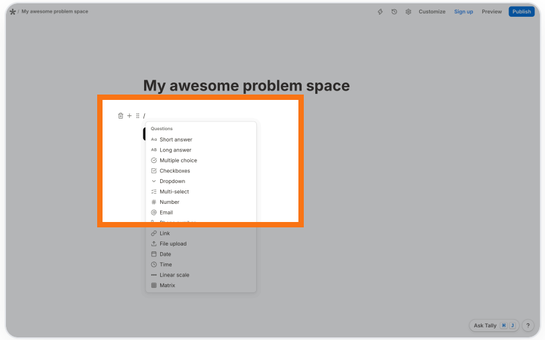
1. **"How often did it hurt last month?"**
   ~~"Tell me about instances when it hurts."~~
2. **Never mention your solution**
   ~~"This is what we build, do you like it?"~~
3. **Role and context, to count your segment**
4. **Short, in their words**



HOW-TO · SURVEY · STEP 1 OF 4

Build the form in ten minutes





1 **Build the form**

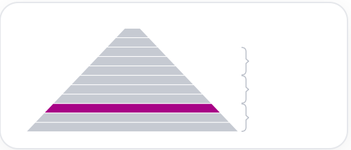
2 Ask about the past

3 Send it to people who acted

4 Let AI count, then check

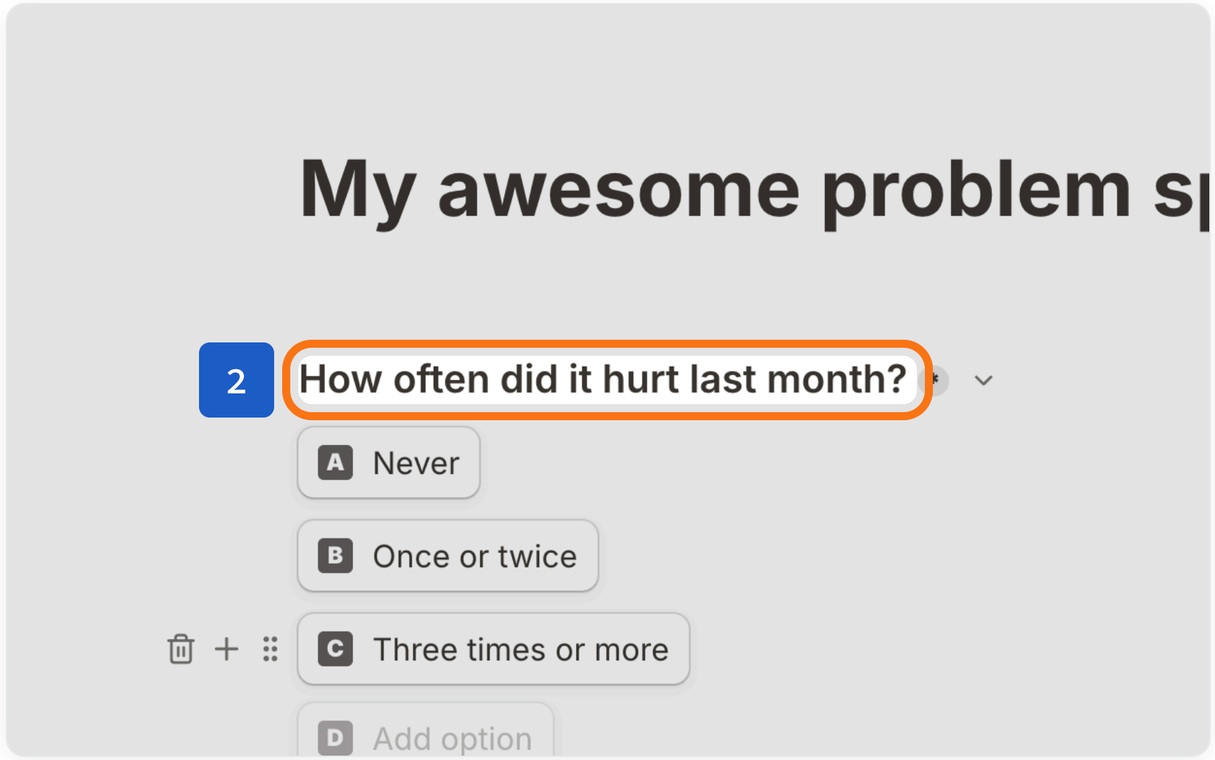
Closed questions only.

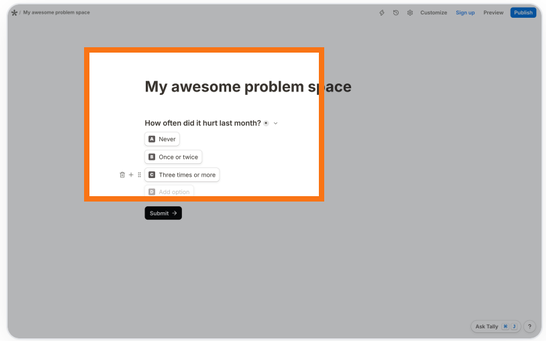
Screenshots of tally.so as of 26 October 2026, captured with scripts/shots.py. Example form typed for this deck. Any form tool works.



HOW-TO · SURVEY · STEP 2 OF 4

Ask about the past

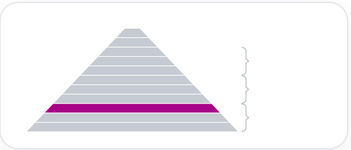




1. Build the form
2. **Ask about the past**
3. Send it to people who acted
4. Let AI count, then check

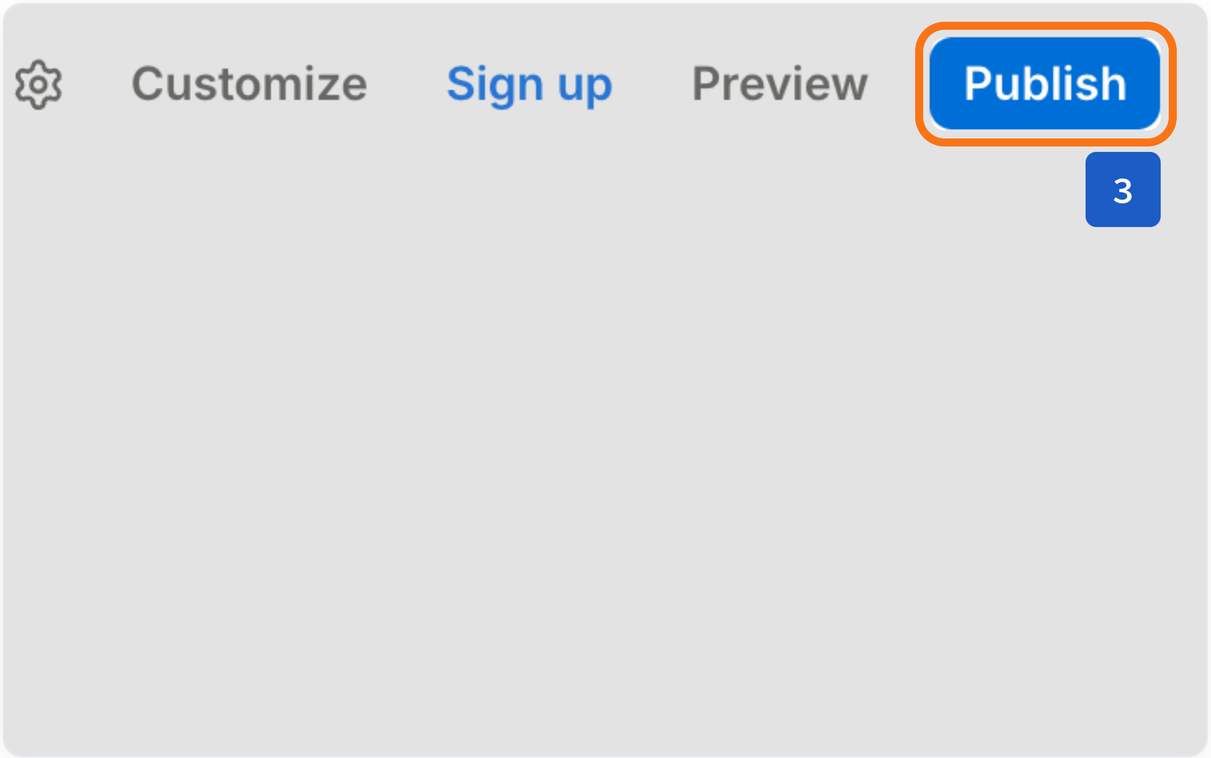
One question about last month, three closed answers

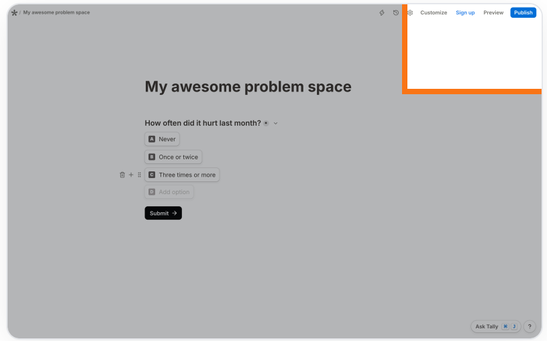
Screenshots of tally.so as of 26 October 2026, captured with scripts/shots.py. Example form typed for this deck. Any form tool works.



HOW-TO · SURVEY · STEP 3 OF 4

Send it to people who acted

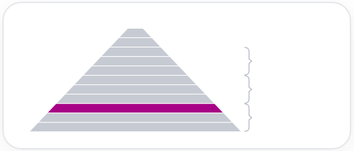




1. Build the form
2. Ask about the past
3. **Send it to people who acted**
4. Let AI count, then check

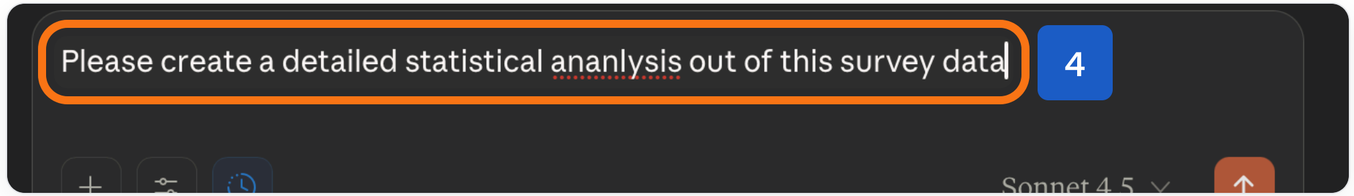
Only to people who already acted, your sign-ups

Screenshots of tally.so as of 26 October 2026, captured with scripts/shots.py. Example form typed for this deck. Any form tool works.



HOW-TO · SURVEY · STEP 4 OF 4

Let AI count. Check every number

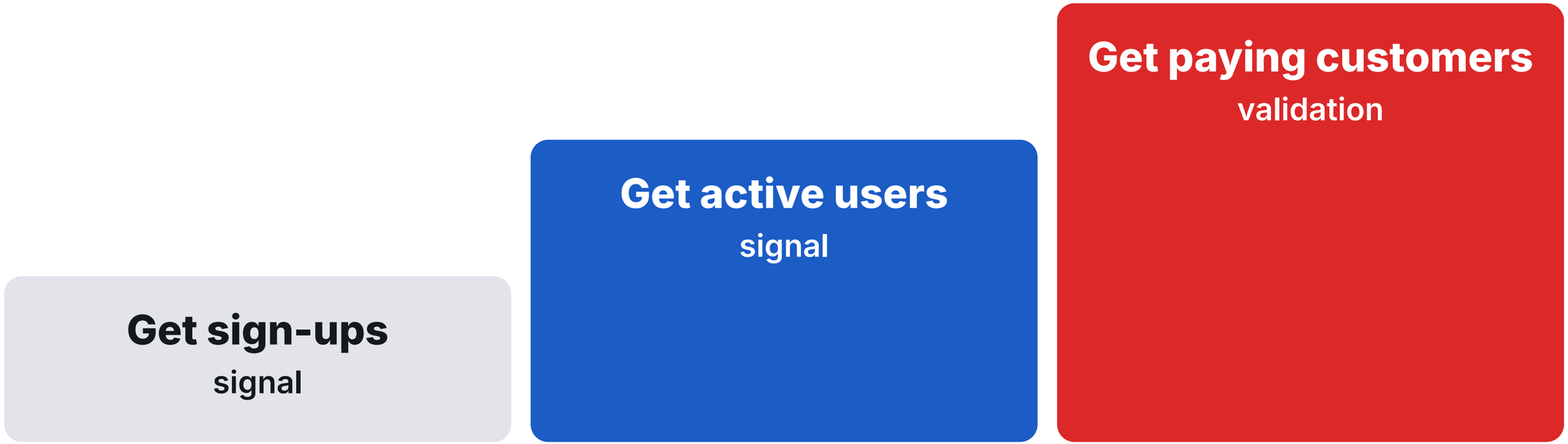


1. Build the form
2. Ask about the past
3. Send it to people who acted
4. **Let AI count, then check**

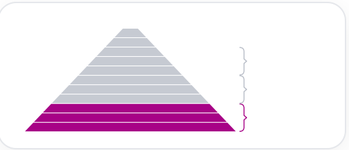
Export without names and emails.

MANTRA

Real validation comes through the credit card



BASTIAN Signal and validation as defined in the course Glossary.



NOW YOU · AS A TEAM · 3 MINUTES

Put your evidence on the pyramid

Anecdote What did one person say?

Interviews What did many say?

Number What did 20+ do?

Limit What does it not prove?

3:00

OUTPUT

Your top step, honestly

INPUT 1 OF 4 · RANK IT · THE ANSWERS

Rank it: saying, doing, paying

1

Which evidence is strongest?

What people do and pay beats what they say. Numbers verify

2

When does a survey count?

Only when it asks about the past and never mentions your solution

3

What validates a problem?

Sign-ups and users are signal. Validation comes through the credit card

INPUT 2 OF 4 · CHAPTER 4 OF 9

Draw it: the line before the data

Write down what would change your mind before the first visitor arrives.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Rank it: what your evidence proves~~

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

INPUT 2 OF 4 · DRAW IT · THREE QUESTIONS

What would change your mind?

1

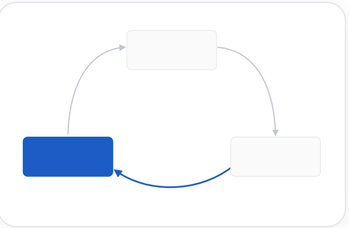
Why write the line before the data?

2

What goes on a test card?

3

What if you miss the line?



WHY IT MATTERS

Decide after the data, and everything says yes

96% · standard psychology papers

positive

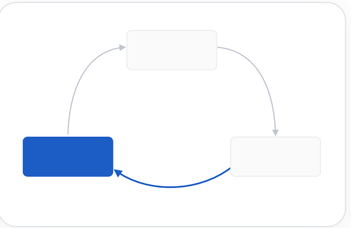
44% · Registered Reports, hypotheses filed first

positive

57% → 8%
heart trials with a benefit, before and after 2000

87%
of 67 top-journal trials needed a correction for their outcomes

RESEARCH Scheel, Schijen & Lakens (2021), 152 standard vs 71 Registered Reports, bars to scale. Kaplan & Irvin (2015), 55 trials. Goldacre et al. (2019).



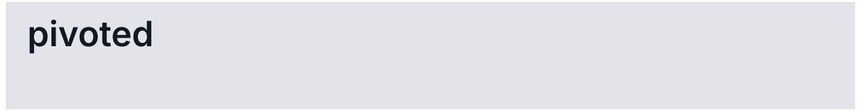
WHY IT MATTERS · FOUNDERS

Founders taught to test pivot more

32% · trained to test hypotheses, 19 of 59



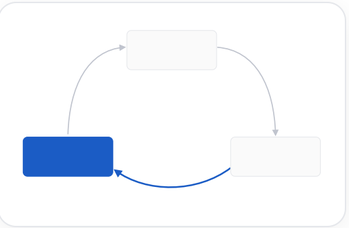
19% · control group, 11 of 57



116
Italian startups, randomized

759
firms in 2024: more ideas stopped

RESEARCH Camuffo, Cordova, Gambardella & Spina (2020), *Management Science*. Camuffo et al. (2024), *Strategic Management Journal*. Bars to scale, 0 to 50%.

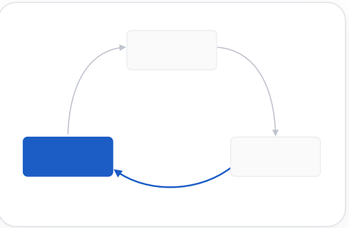


TEST CARD · BEFORE ANYONE VISITS

Write the test before the page exists

We believe	[segment] will [action] when we promise [outcome]
To verify, we	publish the page, bring [N] visitors from [sources]
And measure	the share of segment visitors who take the action
We are right if	[X]% or more by [date], logged today
Then	pass: [next step] · fail: switch segment or problem

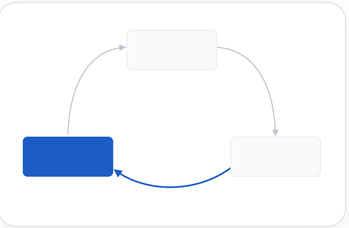
FRAMEWORK Test card after Bland, D. J., & Osterwalder, A. (2019). *Testing Business Ideas*. Wiley. Course material, redrawn with a "Then" row.



EXAMPLE · A TEST CARD

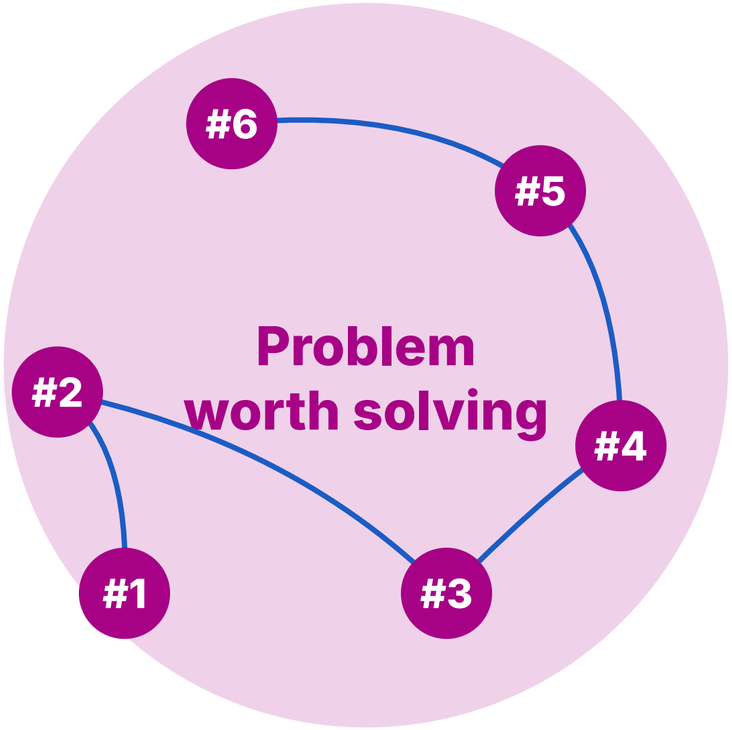
Even an interview test gets a number

We believe	new founders struggle to structure their processes
To verify, we	hold 15 interviews with early-stage founders
And measure	their difficulty, rated from 1 to 10
We are right if	10 out of 15 rate it 7 or higher
Time	2 weeks



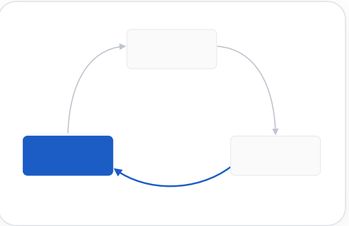
WHEN YOU MISS THE LINE

Missed the line? Move inside the problem space



You are passionate about it
Your audience is passionate about it
You can make a good living from it

Each hop is the next segment or problem



NOW YOU · AS A TEAM · 3 MINUTES

Write your test card

We believe [segment] will [action]

We measure share of segment visitors

We pass at [X]% by [date]

Then pass, and fail

3:00

OUTPUT

One card, dated today

INPUT 2 OF 4 · DRAW IT · THE ANSWERS

Draw it: the line comes first

1

Why write the line before the data?

Decide after the data, and everything says yes. Founders who test pivot more

2

What goes on a test card?

Segment, action, promise, visitors and sources, and the share that passes by a date

3

What if you miss the line?

Move inside the problem space: the next segment or problem

INPUT 3 OF 4 · CHAPTER 5 OF 9

Launch it: a page that tests your problem

A page in their words with one button is your first quantitative test.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Rank it: what your evidence proves~~

04 ~~Draw it: the line before the data~~

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

INPUT 3 OF 4 · LAUNCH IT · THREE QUESTIONS

How does a page test a problem?

1

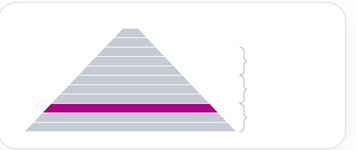
Why a page before a product?

2

What goes on the page?

3

How fast can you build it?

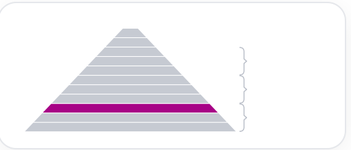


STORY · ZAPPOS, 1999

He sold the shoes before he owned them

1. **No shop has the boots he wants**
2. **He photographs a shop's shoes**
3. **He puts the photos online**
4. **Someone orders. He buys the pair at full price**

DATA Swinmurn, N., as told to D. Eng (2012, 5 September). *Fortune*. Prather, M. (2000, 1 December). Sole proprietor. *Entrepreneur*.



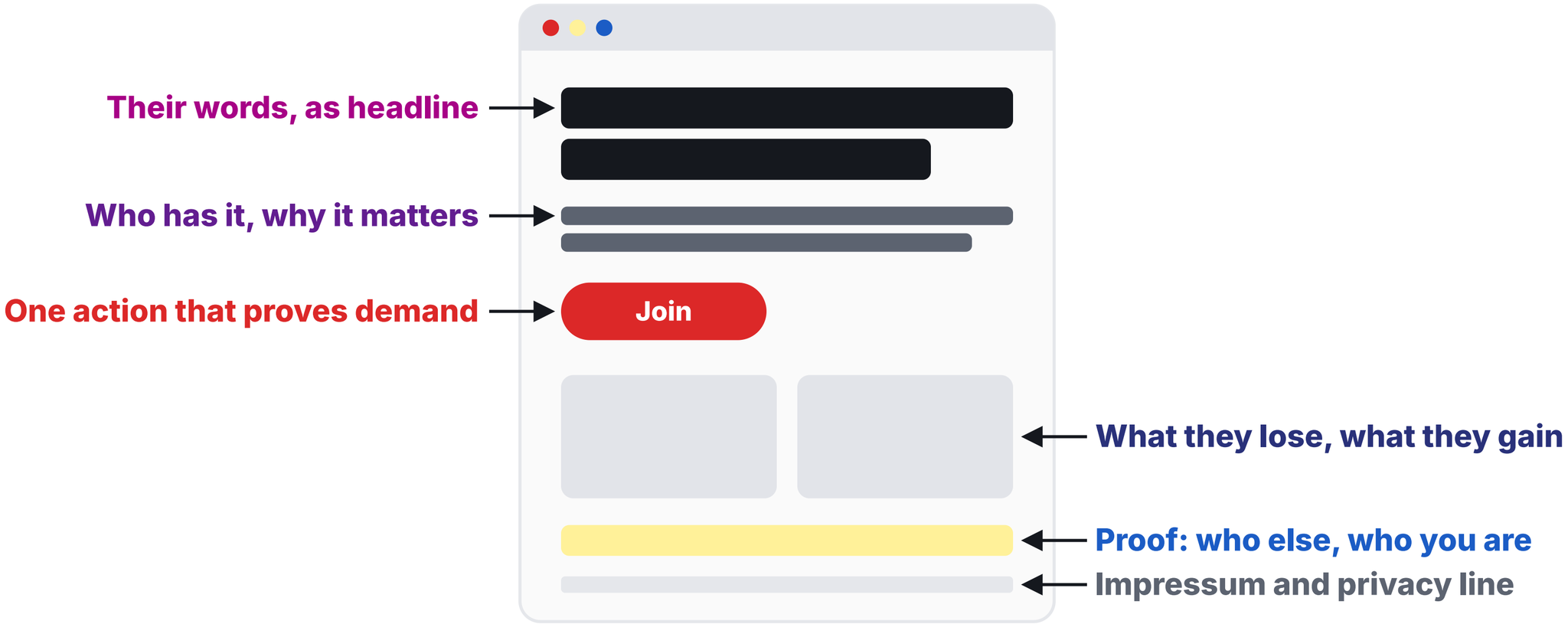
> # The first thing we always tell founders is to launch their product right away …

Geoff Ralston, Y Combinator, 2017

YC Ralston, G. (2017, September 25). YC's essential startup advice. Y Combinator blog.

ANATOMY · PROBLEM-SPACE PAGE

Their problem, their words, one button



A REAL PAGE · READ IT TOP DOWN

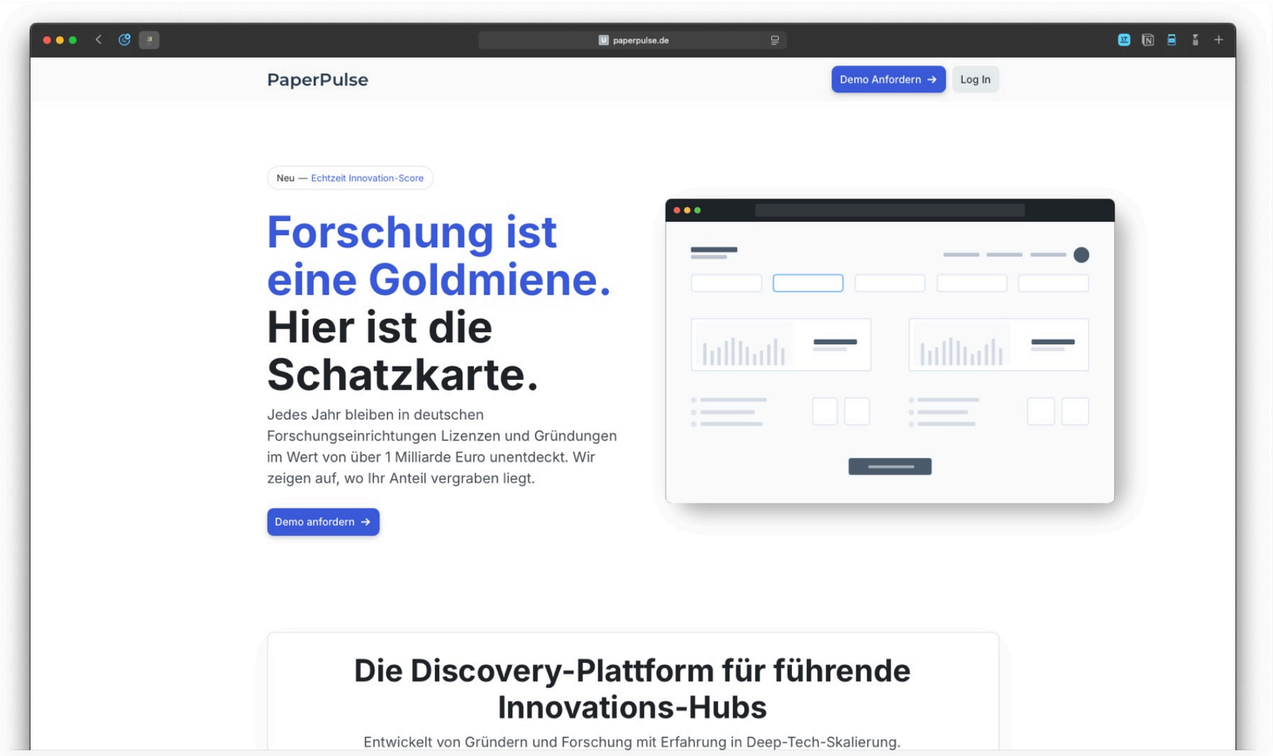
Problem first. The product comes later

HEADLINE

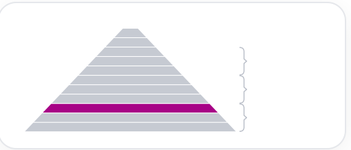
"Research is a gold mine. Here is the treasure map."

ONE ACTION

Request a demo



BASTIAN PaperPulse, Bastian's project, landing page, captured by March 2026. Headline translated from German.



LANDING PAGE CONTENT · TIP 1 OF 4

Lead with the outcome they want

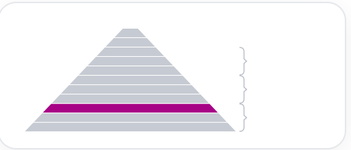
FROM

A hands-on online course

↓

TO

Structure your startup operations



LANDING PAGE CONTENT · TIP 2 OF 4

One line says what, for whom, which change

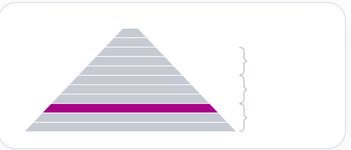
FROM

Bundle is a course

↓

TO

A course for entrepreneurs to structure internal operations



LANDING PAGE CONTENT · TIP 3 OF 4

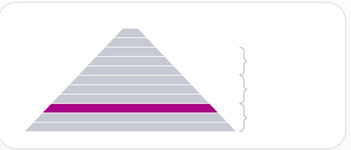
Benefits, not features

FROM

Features of your product

TO

More structure. Less manual work.



LANDING PAGE CONTENT · TIP 4 OF 4

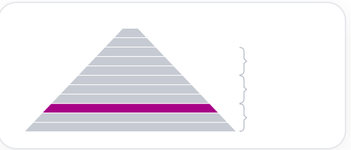
Ask for the action that proves demand

FROM

Learn more

TO

Apply now · Book a call · Buy

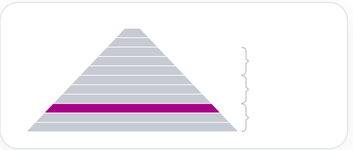


VOTE · HANDS

Which button proves the most demand?

1. **Follow us**
2. **Join the waitlist**
3. **Book a 15-minute call**
4. **Pre-order**

1:00

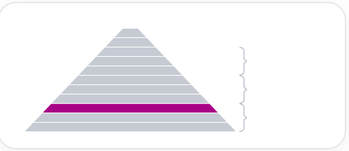


AI COPYWRITING

Don't write from scratch. Start from their words

```
Act as a senior copywriter. Based on these interview notes [PASTE NOTES], write 5 variations of a Landing Page Hero Section (Headline + Subheadline + CTA) that addresses the core pain point directly.
```

A starting point, not the final copy.



NOW YOU · ALONE · 3 MINUTES

Your headline, from one quote

Quote — What did they say, word for word?

Problem — In their words

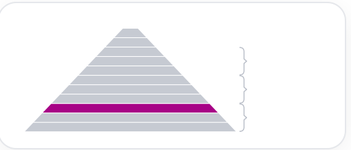
Outcome — What changes for them?

Button — One action

3:00

OUTPUT

One hero section



TOOLING TO TEST A PROBLEM SPACE

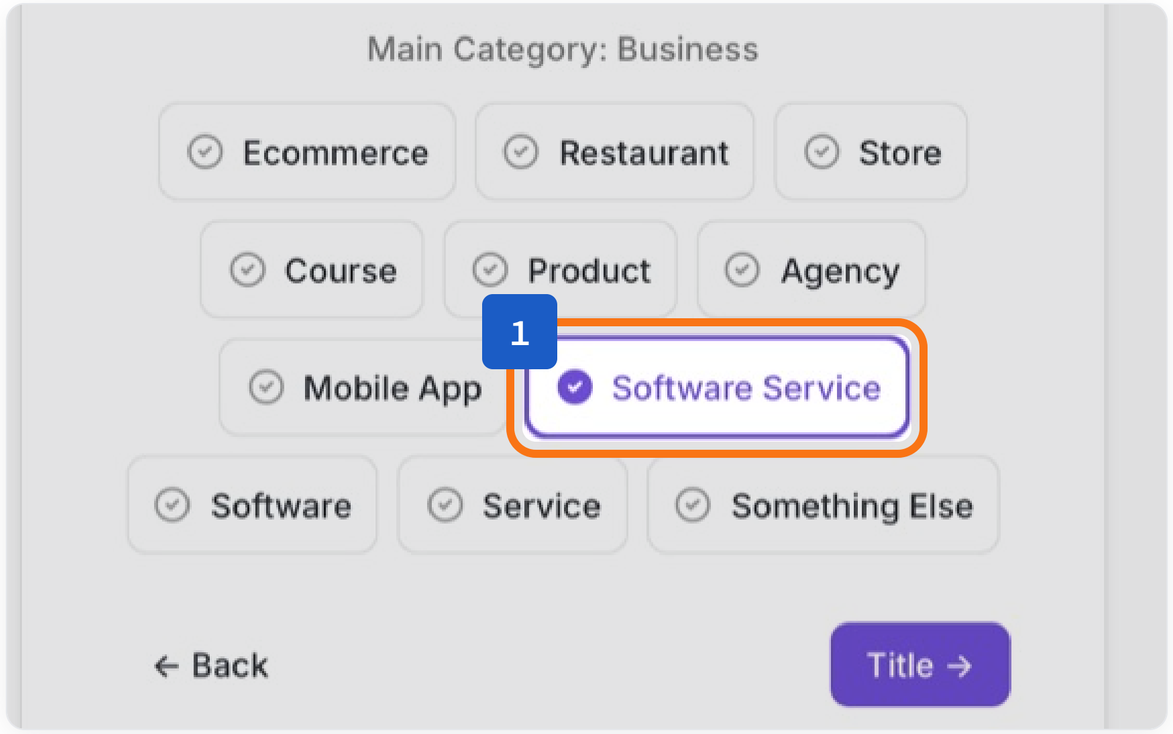
Any builder works. Ship today

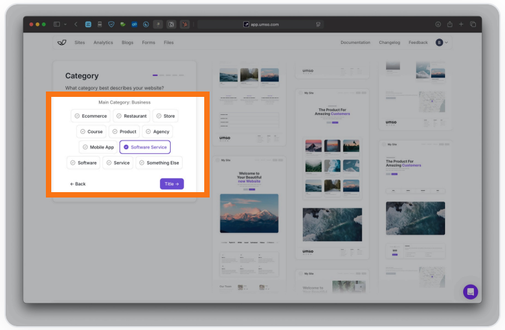
Fast and simple → **Flexible**

Notion	Framer	Webflow	Lovable
A doc as a page	Free plan on a Framer domain	Free starter, more design	Vibe-coded, free daily credits

HOW-TO · BUILD A PAGE · STEP 1 OF 4

Pick a template. Skip the design debate





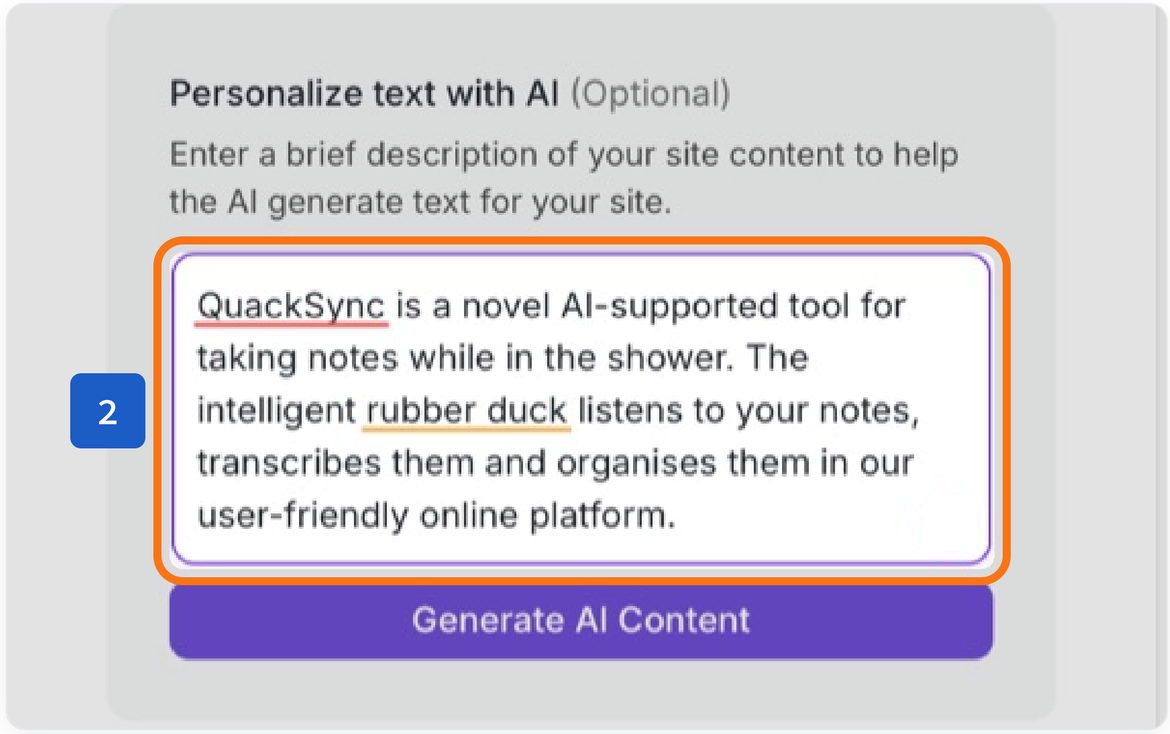
1. **Pick a template**
2. Describe the problem, let AI draft
3. Rewrite in their words
4. Publish on a free subdomain

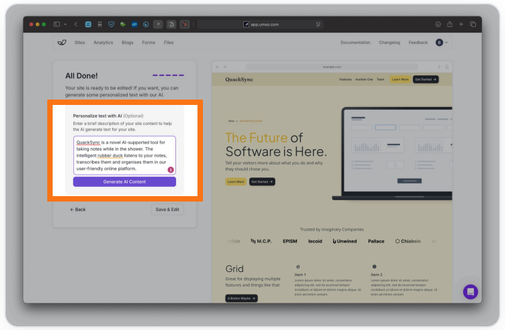
Business, then your category.

BASTIAN Umso on Bastian's screen, 2025, cropped. QuackSync is a made-up example.

HOW-TO · BUILD A PAGE · STEP 2 OF 4

Describe the problem. Let AI draft

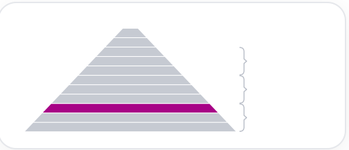




1. Pick a template
2. **Describe the problem, let AI draft**
3. Rewrite in their words
4. Publish on a free subdomain

AI drafts. You decide.

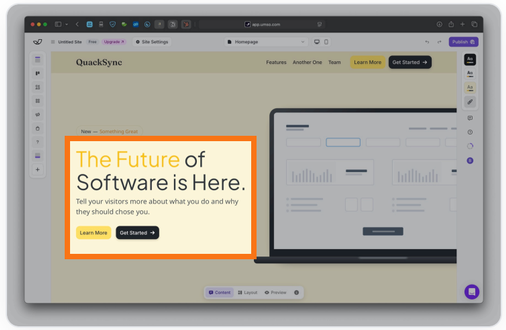
BASTIAN Umso on Bastian's screen, 2025, cropped. QuackSync is a made-up example.



HOW-TO · BUILD A PAGE · STEP 3 OF 4

Rewrite the hero in their words





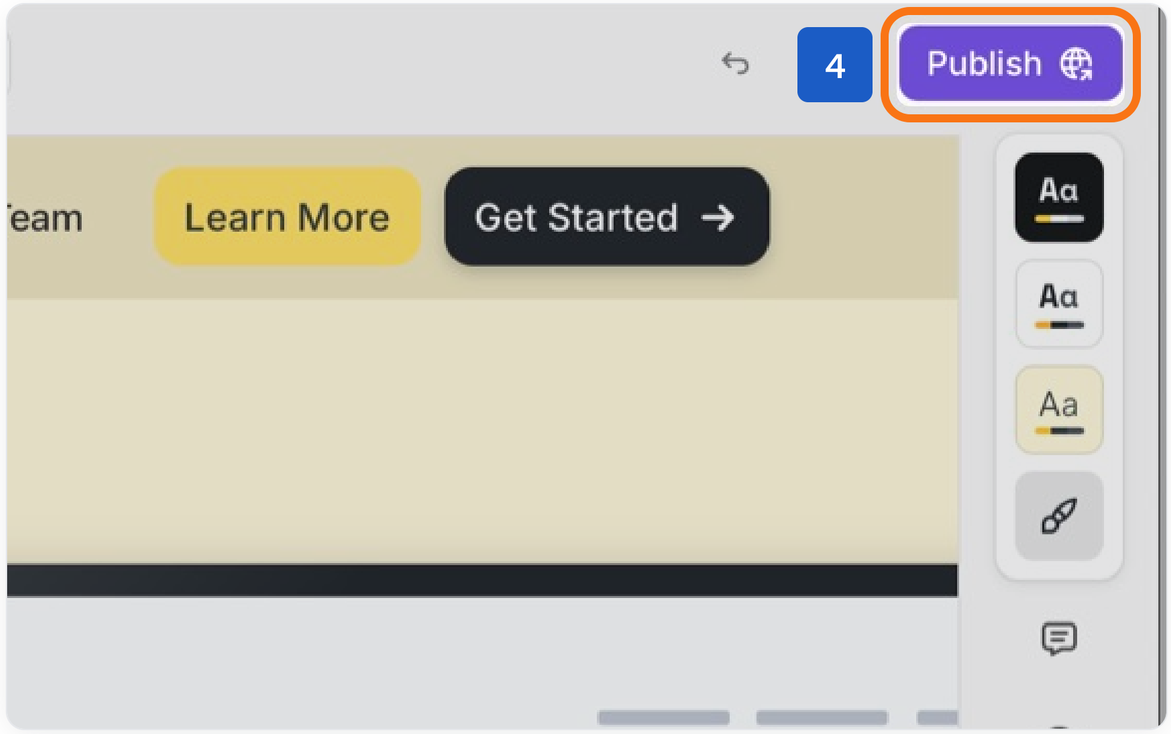
1. Pick a template
2. Describe the problem, let AI draft
3. **Rewrite in their words**
4. Publish on a free subdomain

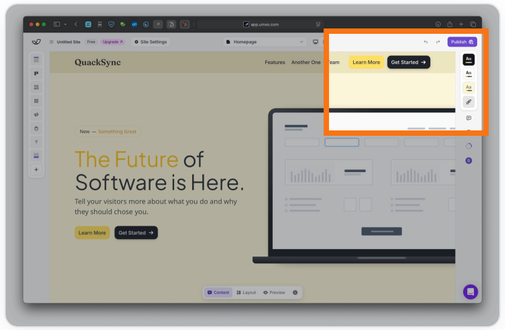
Their sentence goes here.

BASTIAN Umso on Bastian's screen, 2025, cropped. QuackSync is a made-up example.

HOW-TO · BUILD A PAGE · STEP 4 OF 4

Publish on a free subdomain





1. Pick a template
2. Describe the problem, let AI draft
3. Rewrite in their words
4. **Publish on a free subdomain**

Domains wait until the page earns one

BASTIAN Umso on Bastian's screen, 2025, cropped. QuackSync is a made-up example.

INPUT 3 OF 4 · LAUNCH IT · THE ANSWERS

Launch it: their words, one button

1

Why a page before a product?

Sell before you own it, as Zappos did. Launch right away

2

What goes on the page?

Their problem in their words, the outcome they want, one button that proves demand

3

How fast can you build it?

Today. Any builder works. AI drafts, you rewrite the hero in their words

INPUT 4 OF 4 · CHAPTER 6 OF 9

Count it: visitors from your segment

Bring the people who have the problem, and count only them.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Rank it: what your evidence proves~~

04 ~~Draw it: the line before the data~~

05 ~~Launch it: a page that tests your problem~~

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

INPUT 4 OF 4 · COUNT IT · THREE QUESTIONS

Who comes, and who counts?

1

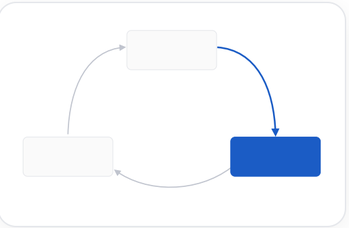
How many visitors do you need?

2

Where do you find them?

3

How do you read the result?

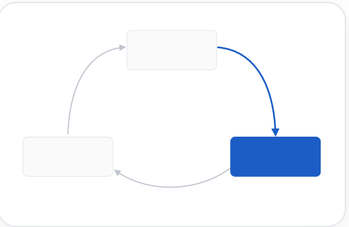


LET'S TALK CONVERSION

Work backwards from the sign-ups you need

Impressions	Clicks	Sign-ups
740	25	1
	3.4% click-through	4% sign-up

Work backwards: how many impressions for the sign-ups you need?



SPEED MATTERS

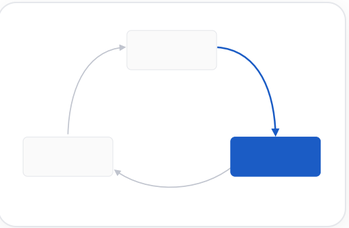
To get, every day, each of you, at least

15 visitors

```
  300 see your link
× 5%  click → 15
  × 7 days → 105
------------------
× 10% sign up → 10
```

300 a day see your link!

ILLUSTRATIVE Arithmetic, redone per student from the homework bar (100 visitors per student) and the default line (10%). The 5% click-through is an assumption, not a benchmark.

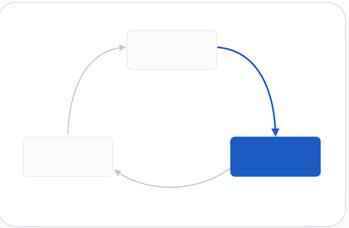


DISTRIBUTION

Go where your segment already is

1% inspiration ~~99% perspiration~~ **99% distribution**

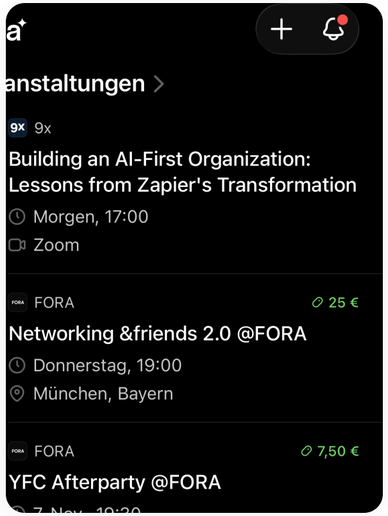
- **B2B**
  - LinkedIn
  - Industry groups
- **Developer tools**
  - Hacker News
  - Product Hunt
- **B2C**
  - Instagram
  - WhatsApp
  - X
- **Niche audience**
  - Discord
  - Slack
- **Niche problem**
  - Facebook groups
  - Forums



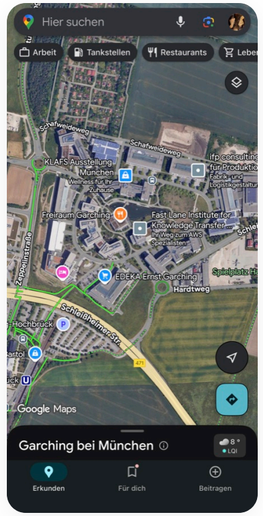
FINDING THE ROOM

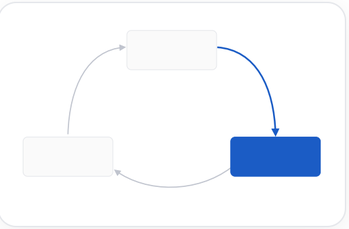
Name the room, not the channel

ONLINE COMMUNITIES · ONE NAMED LUMA EVENT



OFFLINE PLACES · A CAMPUS CANTEEN

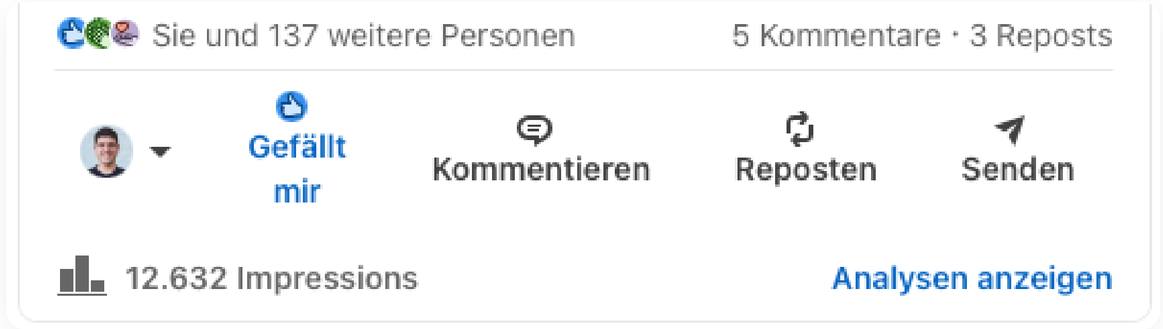




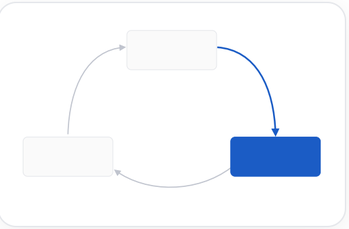
BUILDING AN AUDIENCE

Post what you learn. Talk to who reacts

1. **Post what you found, what surprised you**
2. **Message everyone who reacts**
3. **Check: are they your segment?**

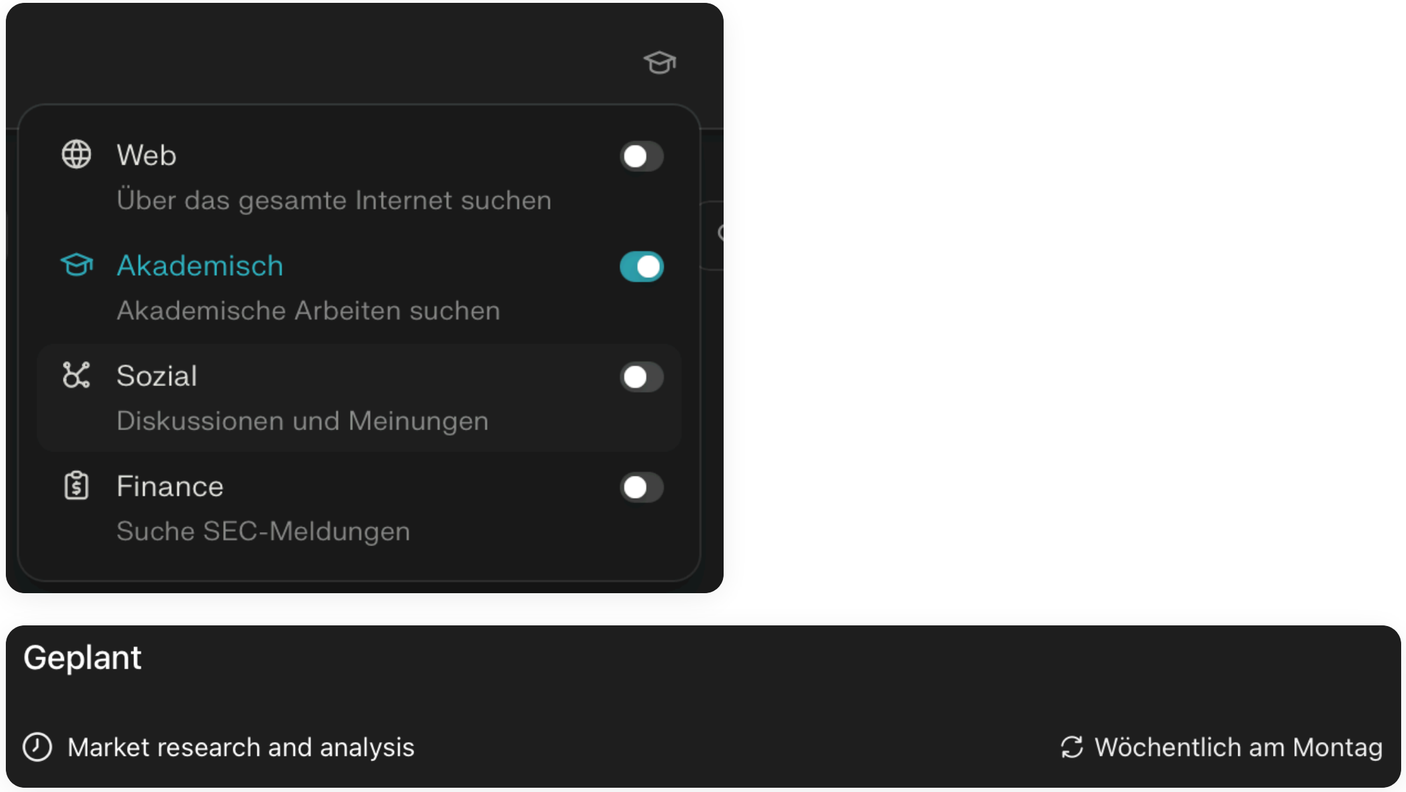


BASTIAN Building an audience. LinkedIn statistics of Bastian's post, captured by March 2026.



HOW-TO · LET AI DO THE HEAVY LIFTING

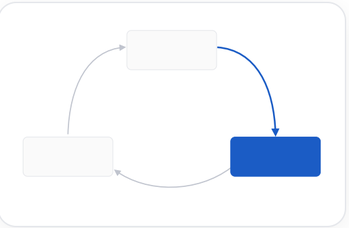
Let an agent search for you, every Monday



1. **Ask where your segment talks**
2. **Schedule it weekly**
3. **Open every source**

Leads end in a named person.

BASTIAN Perplexity sources and a ChatGPT scheduled task on Bastian's screen, 2025, cropped.



BENCHMARK · WARM AND COLD

Warm traffic converts. Cold traffic costs

16.9% · email, people on your own list

warm

4.1% · paid search

2.9% · paid social

3.8% · all SaaS landing pages

DATA Unbounce, Conversion Benchmark Report 2024, SaaS medians. A conversion is any click or form event. Bars to scale, 0 to 20%.

COLD TRAFFIC · BASTIAN'S OWN ACCOUNT

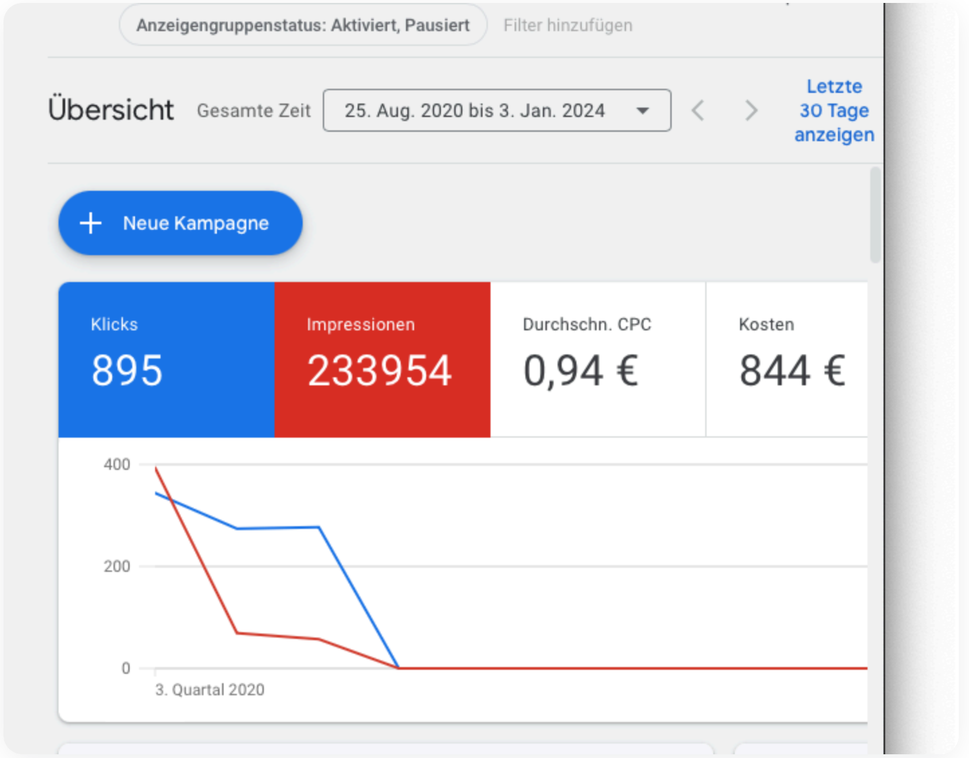
Cold clicks cost €0.94 each

233,954
impressions

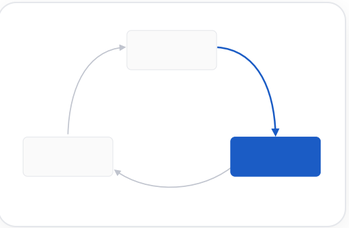
895
clicks

0.38%
clicked

€844
spent



BASTIAN Google Ads account overview, 25 August 2020 to 3 January 2024. Click rate: 895 ÷ 233,954.



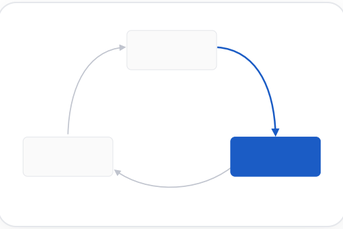
TAGGED LINKS

Tag every link, or you can't tell who converts

```
yourpage.de/?utm_source=linkedin&utm_medium=post&utm_campaign=m04&utm_content=anna
```

Where | **How** | **Which test** | **Who sent it**

DATA Google, Analytics Help, URL builders: collect campaign data with custom URLs, as of October 2026. Example link illustrative.



HOW-TO · ANALYTICS · STEP 1 OF 4

Set up the dashboard before the first visitor

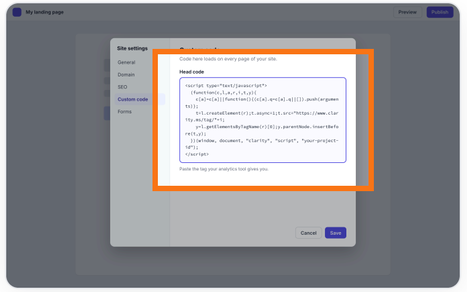
Code here loads on every page of your site.

Head code

1

```
<script type="text/javascript">
  (function(c,l,a,r,i,t,y){
    c[a]=c[a]||function(){(c[a].q=c[a].q||[]).push(argumen
ts)};
    t=l.createElement(r);t.async=1;t.src="https://www.clar
ity.ms/tag/"+i;
    y=l.getElementsByTagName(r)[0];y.parentNode.insertBefo
re(t,y);
  })(window, document, "clarity", "script", "your-project-
id");
</script>
```

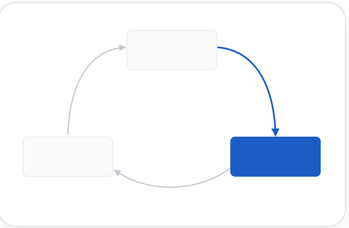
Paste the tag your analytics tool gives you.



1. **Add the tag to your page**
2. Add a consent banner for EU visitors
3. Filter by source
4. Watch your segment's sessions

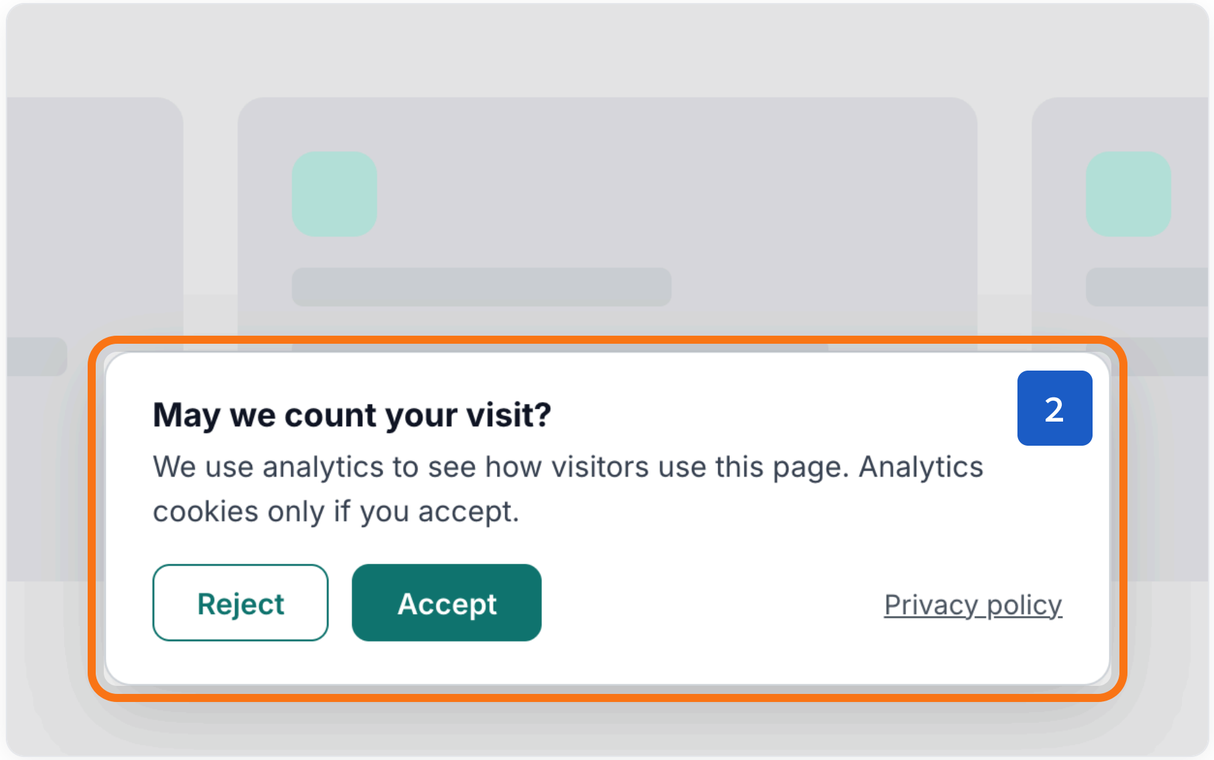
In Clarity, Settings, then Setup, then Install manually

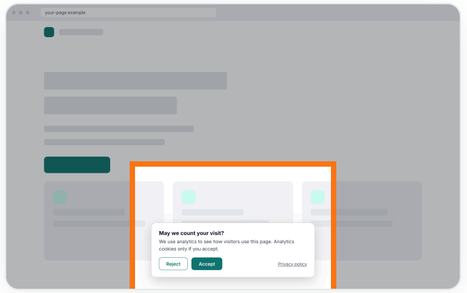
ILLUSTRATIVE Screen prototyped for this deck. Setup path from the Microsoft Clarity documentation, as of October 2026.



HOW-TO · ANALYTICS · STEP 2 OF 4

Add a consent banner for EU visitors

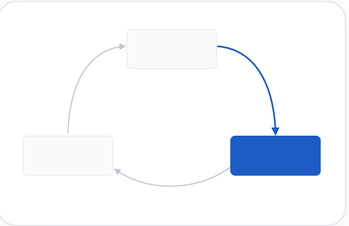




1. Add the tag to your page
2. **Add a consent banner for EU visitors**
3. Filter by source
4. Watch your segment's sessions

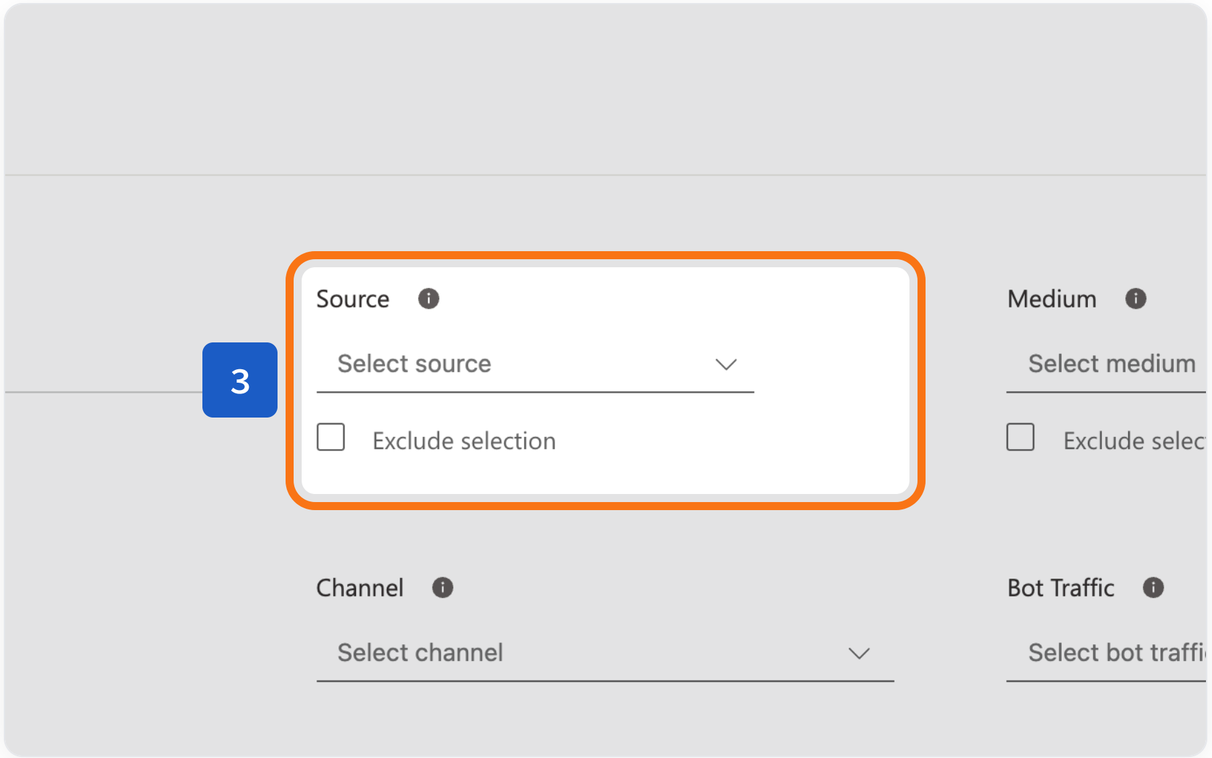
EU visitors: banner first.

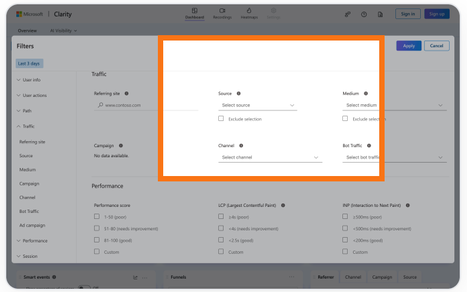
ILLUSTRATIVE Screen prototyped for this deck. Consent rules from the Microsoft Clarity documentation, as of October 2026.



HOW-TO · ANALYTICS · STEP 3 OF 4

Filter by source

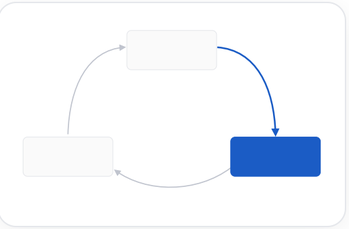




1. Add the tag to your page
2. Add a consent banner for EU visitors
3. **Filter by source**
4. Watch your segment's sessions

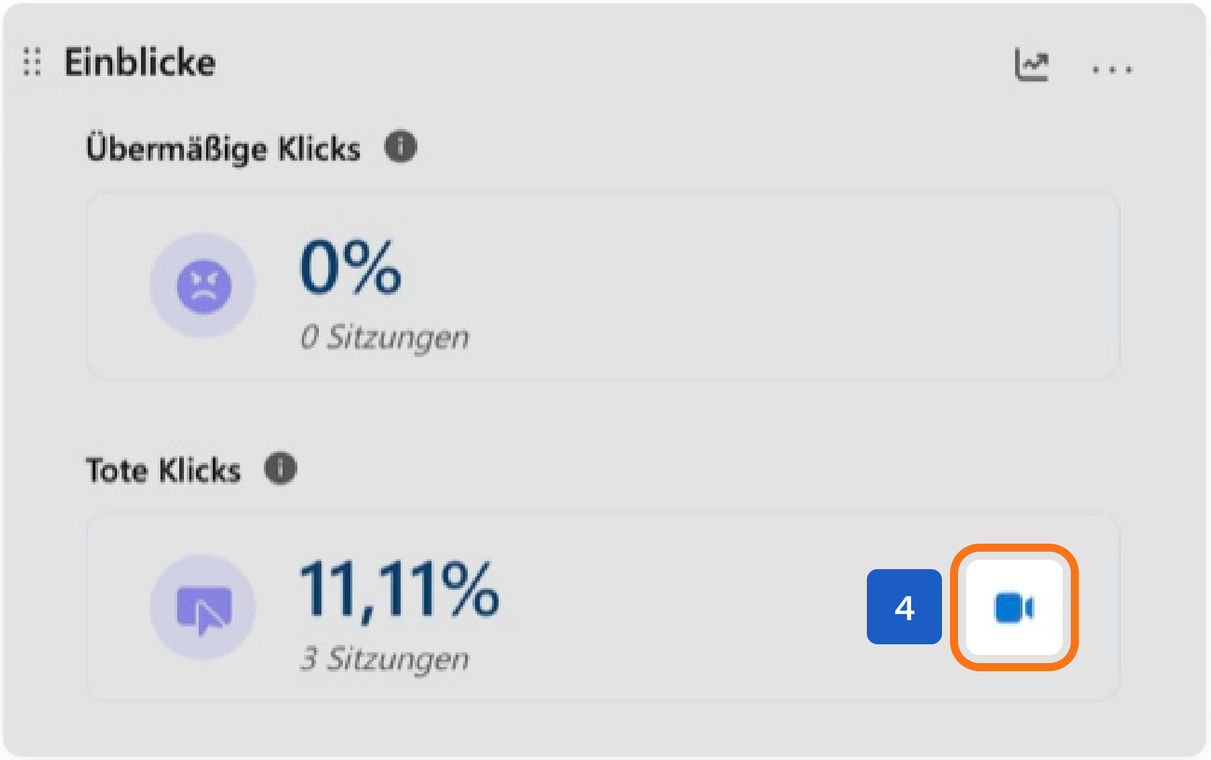
Filters, then Traffic, then Source

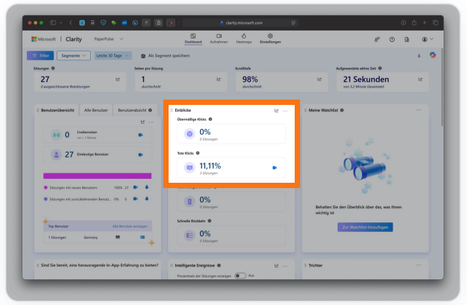
Screenshots of Microsoft Clarity's public live demo as of 26 October 2026, captured with scripts/shots.py.



HOW-TO · ANALYTICS · STEP 4 OF 4

Watch your segment's sessions

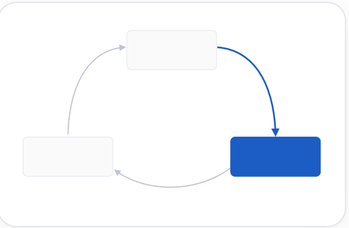




1. Add the tag to your page
2. Add a consent banner for EU visitors
3. Filter by source
4. **Watch your segment's sessions**

Tote Klicks are dead clicks, 11% of sessions

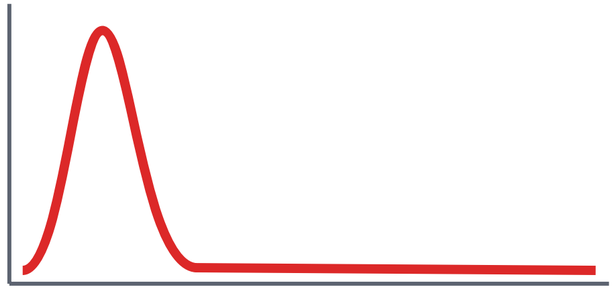
BASTIAN Microsoft Clarity on Bastian's screen, PaperPulse, captured by March 2026, cropped.



READING YOUR RESULT

Read the curve, then your number

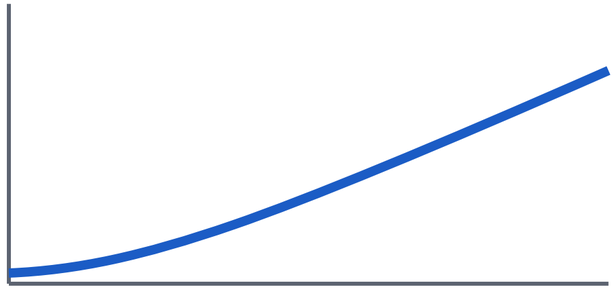
No interest



Wrong group, problem or words

Rework the problem space

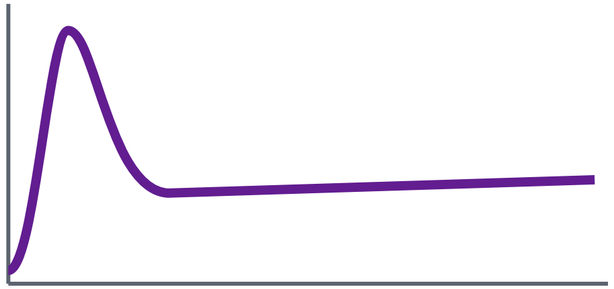
Organic interest



Right group, sustained interest, good words

Find more of them

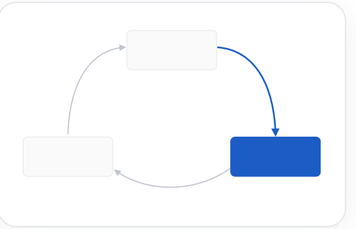
External shock



An article or a post went viral

Don't count the spike

Impressions < clicks < sign-ups. Remember the hierarchy of evidence!



DOUBLE DOWN · AN EXAMPLE

Double down where your segment converts

11.5% · LinkedIn posts in 2 wind-industry groups

7 of 61

16.7% · Direct outreach, 30 emails

4 of 24

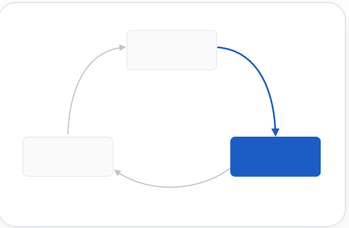
2.3% · €50 search ads

2 of 87

12.9%
segment only, 11 of 85

33
visitors from friends and uni, excluded

ILLUSTRATIVE Synthetic course exemplar, Landing-Page Test, team TurbineTrack. Bars to scale, 0 to 20%. Double down.



NOW YOU · AS A TEAM · 3 MINUTES

Your traffic math

Sign-ups How many do you want?

Share What do you expect?

Visitors So how many?

Places Name three, with people

3:00

OUTPUT

Three named places

INPUT 4 OF 4 · COUNT IT · THE ANSWERS

Count it: your segment, by source

1

How many visitors do you need?

Work backwards from the sign-ups you need: at least 15 a day each

2

Where do you find them?

Where your segment already is. Name the room, not the channel

3

How do you read the result?

Tag every link, count only your segment, read the curve, then your number

CHAPTER 7 OF 9

Wrap-up and check-out

What you take into the week, and how today went for you.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Rank it: what your evidence proves~~

04 ~~Draw it: the line before the data~~

05 ~~Launch it: a page that tests your problem~~

06 ~~Count it: visitors from your segment~~

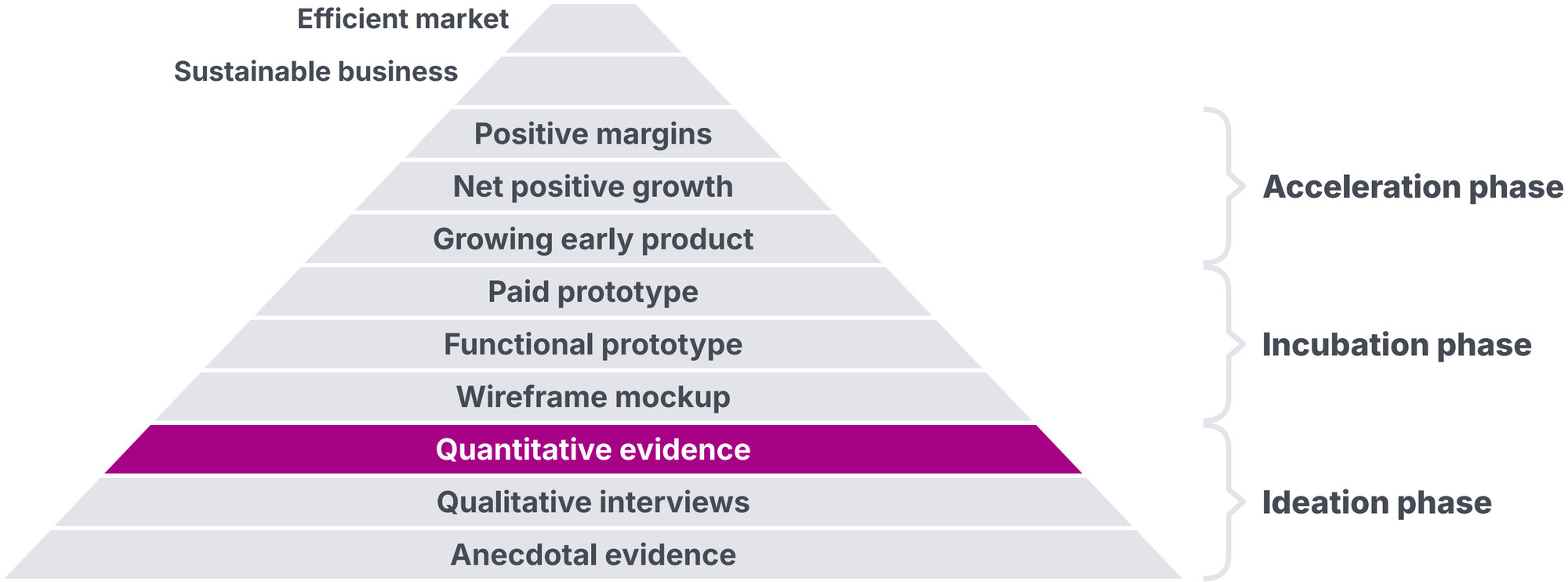
07 **Wrap-up and check-out**

08 Workshop: launch a page that tests your problem

09 Case discussion

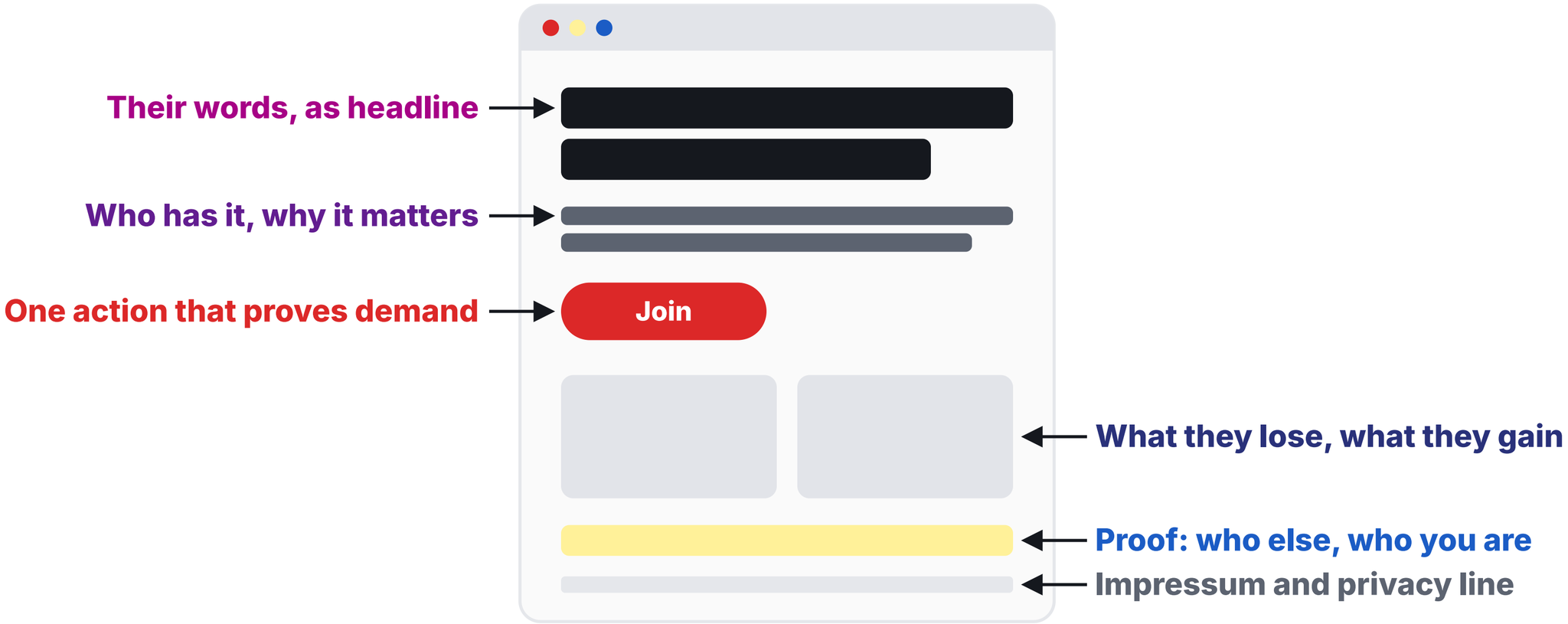
RECAP · STEP THREE IS A SURVEY OR A PAGE, 20+ PEOPLE

From today you climb from saying to doing



RECAP · YOUR PAGE

Their problem, their words, one button



REFLECTION 1 OF 3 · THE CONTENT

What could you apply?



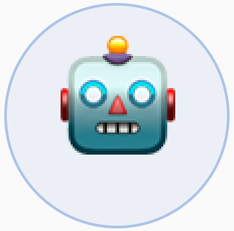
What could you apply?



What gave you a hard time?

REFLECTION 2 OF 3 · THE PROCESS

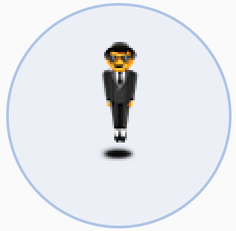
Man vs. machine



Where did AI actually help you?



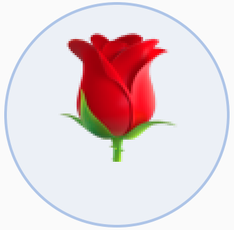
Where was your genius needed?



What are your best hacks to share?

REFLECTION 3 OF 3 · THE RESULTS

Rose, thorn, bud



What was the breakthrough today?



What was the hardest part today?



What is an idea/feature you didn't have time to finish, but want to explore later?

HOMEWORK UNTIL NEXT TIME

Publish one landing page per team, log its pass or fail line before the first visitor arrives, and bring 100 visitors per student from your segment to it.

100 each

The learning floor is the live page, the logged line and 30 visitors per student from your segment.

Your line is logged before anyone visits.

HOMEWORK · HOW TO DO IT

Seven moves for the week

1. **Start with your Problem space (M03) prospects**
2. **Go where your segment is**
3. **Tag every link**
4. **Never move the line, log any copy change**
5. **Call three sign-ups each**
6. **Survey your sign-ups, optionally**
7. **Get a blind second opinion**

HOMEWORK · WHAT GOOD LOOKS LIKE

A good week ends on your line, not on a feeling

	GOOD	WEAK
Line	10%, logged before launch	Written after the first numbers
Traffic	140 tagged visitors from two recruiter groups	400 visitors, mostly friends
Result	9 of 140, under the line, segment switched	"Lots of interest!"

ILLUSTRATIVE Made-up example from the Validating problems (M04) homework, the recruiting team of the workshop.

Sources in this deck

Bland, D. J., & Osterwalder, A. (2019). *Testing business ideas*. Wiley.

Bradburn, N. M., Rips, L. J., & Shevell, S. K. (1987). Answering autobiographical questions. *Science*, 236(4798), 157-161.

Camuffo, A., Cordova, A., Gambardella, A., & Spina, C. (2020). A scientific approach to entrepreneurial decision making. *Management Science*, 66(2), 564-586.

Camuffo, A., et al. (2024). A scientific approach to entrepreneurial decision-making: Large-scale replication and extension. *Strategic Management Journal*, 45(6).

Goldacre, B., et al. (2019). COMPare: A prospective cohort study correcting and monitoring 58 misreported trials. *Trials*, 20, 118.

Google. (2026). *URL builders: Collect campaign data with custom URLs*. Analytics Help.

Kaplan, R. M., & Irvin, V. L. (2015). Likelihood of null effects of large NHLBI clinical trials has increased over time. *PLoS ONE*, 10(8).

Sources in this deck

LocaliQ. (2026). *Search advertising benchmarks for every industry*.

McCarthy, A. M., Schoorman, F. D., & Cooper, A. C. (1993). Reinvestment decisions by entrepreneurs. *Journal of Business Venturing*, 8(1), 9-24.

Microsoft. (2026). *Consent management*. Microsoft Clarity documentation.

Microsoft. (2026). *How to set up Clarity manually*. Microsoft Clarity documentation.

Murphy, J. J., et al. (2005). A meta-analysis of hypothetical bias in stated preference valuation. *Environmental & Resource Economics*, 30(3).

NIST/SEMATECH. *e-Handbook of statistical methods*, 7.2.4.1, confidence intervals.

Prather, M. (2000, December 1). Sole proprietor. *Entrepreneur*.

Ralston, G. (2017, September 25). *YC's essential startup advice*. Y Combinator blog.

Sources in this deck

Ries, E. (2011). *The lean startup*. Crown Business.

Scheel, A. M., Schijen, M. R. M. J., & Lakens, D. (2021). An excess of positive results. *Advances in Methods and Practices in Psychological Science*, 4(2).

Schmidt, J., & Bijmolt, T. H. A. (2020). Accurately measuring willingness to pay for consumer goods. *Journal of the Academy of Marketing Science*, 48(3).

Seibel, M. (2023). *How to build an MVP*. Y Combinator Startup Library.

Swinmurn, N., as told to Eng, D. (2012, September 5). Nick Swinmurn: Zappos' silent founder. *Fortune*.

Unbounce. (2024). *Conversion benchmark report 2024*, SaaS.

See you next time!

Workshop: Launch a page that tests your problem

Write the test, launch the page, and send your first visitors before you leave.

What evidence would change your mind?

01 Previously on Ambitious Entrepreneurship

02 Your presentations

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 **Workshop: launch a page that tests your problem**

09 Case discussion

WORKSHOP · STEP 1 OF 3

Write the test, then the copy

1 STEP 1 · 30 MIN

WRITE THE TEST, THEN THE COPY

Decide what result would change your mind

→ We believe [segment] will [action] when we promise [outcome]
→ Kill line in the Kill Criteria Log, dated, before the page exists
→ Raw notes into an AI, names removed, five hero sections
→ Pick the headline closest to what someone said

Output Test card, dated kill line, hero section

Minutes 30

30:00

WORKSHOP · STEP 2 OF 3

Launch in under an hour

2 STEP 2 · 35 MIN

LAUNCH IN UNDER AN HOUR

A live page that names their problem, asks for one action

→ Their words, why it matters, what changes, one button
→ Form: email plus one segment question
→ Impressum and privacy line, or it stays offline
→ Analytics with visitors and sign-ups per source, then publish
→ Five-second test with another team

Output A public link and analytics that count

Minutes 35

35:00

WORKSHOP · STEP 3 OF 3

Bring your segment, then commit

3 STEP 3 · 25 MIN

BRING YOUR SEGMENT, THEN COMMIT

Leave with visitors on their way

→ Work backwards: 10 sign-ups at 10% need 100 visitors
→ Prospects first, then communities, groups, newsletters
→ One tagged link per student and channel
→ First messages sent before you leave
→ Each student picks floor or full bar, midweek pace

Output Tagged links, sent messages, targets, pace

Minutes 25

25:00

ARRIVED EMPTY-HANDED?

Start in the forums where they complain

No interview notes? Copy your segment's exact sentences from the forums where it complains, mark the promise as assumed, and post your link there in Step 3.

HOMEWORK UNTIL NEXT TIME

Publish one landing page per team, log its pass or fail line before the first visitor arrives, and bring 100 visitors per student from your segment to it.

100 each

The learning floor is the live page, the logged line and 30 visitors per student from your segment.

Your line is logged before anyone visits.

See you next time!

MODULE 04 · VALIDATING PROBLEMS · CASE DISCUSSION · ABOUT 90 MINUTES

Case discussion: Fairphone

Stand in Fairphone's shoes on 13 May 2013 and decide whether to sell 5,000 phones before making one.

Do you open the pre-sale on this benchmark?

01 Previously on Ambitious Entrepreneurship

02 Your presentations

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 **Case discussion**

CASE · FAIRPHONE, AMSTERDAM, 2010 TO 13 MAY 2013

Three years of campaign, one month to sell 5,000



DATA Validating problems (M04) case, timeline and Exhibit 1. Nothing after 13 May 2013.

CASE · MONDAY 13 MAY 2013 · VOTE 1 OF 2

Open the pre-sale on this benchmark, or not yet?

OPTION A

Open the pre-sale on this benchmark

OPTION B

Not yet

€325
the price, paid up front

5,000
phones sold

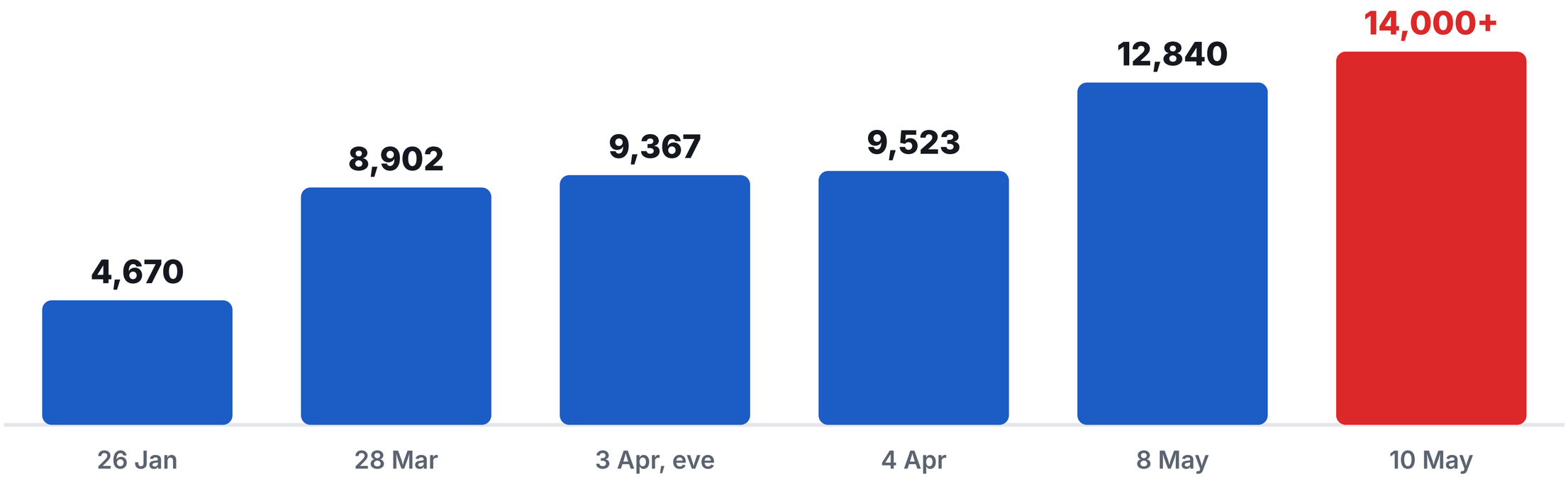
15 June
2013, the deadline

Refund
if the orders are cancelled

DATA Validating problems (M04) case, Exhibit 2 (shop terms, articles 4.3 and 4.4, adopted 13 May 2013) and Exhibit 3.

CASE · EXHIBIT 1 · NEWSLETTER SUBSCRIBERS, 2013

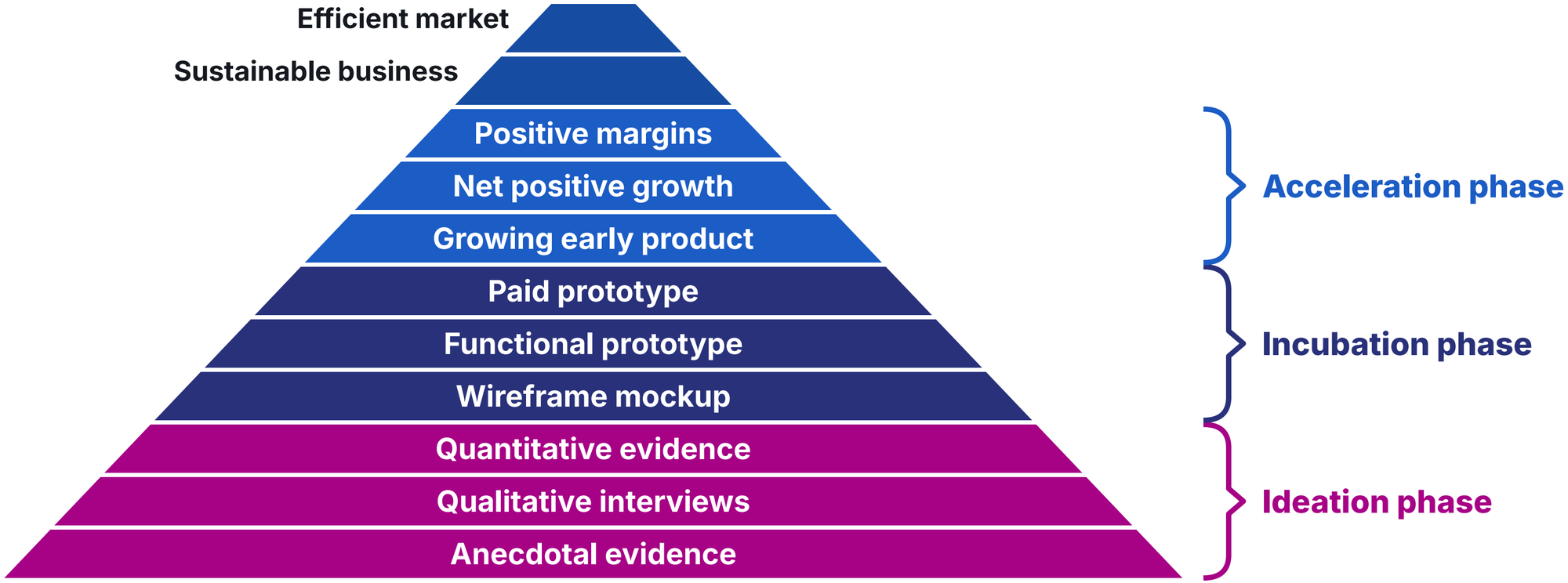
What do 14,000 sign-ups prove?



DATA Validating problems (M04) case, Exhibit 1: archived newsletter counter, Fairphone blog of 4 April 2013 (company figures for the evening of 3 April and for 4 April), Telecompaper of 10 May 2013 ("more than 14,000", drawn at 14,000). Bars to scale, 0 to 15,000.

CASE · DIAGNOSIS

Which step does each piece stand on?



BASTIAN Hierarchy of evidence in the lean startup. Filled on the board from the case.

Fairphone has a sales benchmark of 5,000 (five thousand) mobile phones, which must be sold before the 15th of June 2013.

Shop terms, article 4.3, adopted 13 May 2013. If Fairphone cancels on a miss, the buyer gets "a full refund of the money the Customer already has paid" (4.4)

DATA Validating problems (M04) case, Exhibit 2: Fairphone B.V., Customer Terms and Conditions, articles 4.3 and 4.4, adopted 13 May 2013.

CASE · BOARD PLAN

Is 5,000 by 15 June a good test of demand?

Vote 1	Vote 2	Prediction

Evidence on 13 May and its step	**What it proves** and what it doesn't	**The line** as a test	**5,000 buyers** from where?
What, and which step?	Proves · doesn't prove	Count · threshold · date Pass · miss	Share of the list, then?

CASE · VOTE 2 OF 2

Same question. Who moved, and why?

OPTION A

Open the pre-sale on this benchmark

OPTION B

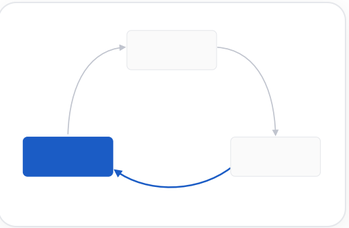
Not yet

€325
the price, paid up front

5,000
phones sold

15 June
2013, the deadline

Refund
if the orders are cancelled



CASE · BRIDGE

Before your page exists, write your line

Fairphone wrote its line into its terms before the first buyer arrived.

We are right if	[X]% or more by [date], logged today
Then	pass: [next step] · fail: switch segment or problem

FRAMEWORK The last two rows of the test card, after Bland and Osterwalder (2019), as on slide 30.

See you next time!